Hurl Documentation

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Introduction



What's Hurl?

Hurl is a command line tool that runs **HTTP requests** defined in a simple **plain text format**.

It can chain requests, capture values and evaluate queries on headers and body response. Hurl is very versatile: it can be used for both **fetching data** and **testing HTTP** sessions.

Hurl makes it easy to work with **HTML** content, **REST / SOAP / GraphQL** APIs, or any other **XML / JSON** based APIs.

```
# Go home and capture token
GET https://example.org
HTTP 200
[Captures]
csrf_token: xpath "string(//meta[@name='_csrf_token']/@content)"


# Do login!
POST https://example.org/login
[Form]
user: toto
password: 1234
token: {{csrf_token}}
HTTP 302
```

Chaining multiple requests is easy:

```
GET https://example.org/api/health
GET https://example.org/api/step1
GET https://example.org/api/step2
GET https://example.org/api/step3
```

Also an HTTP Test Tool

Hurl can run HTTP requests but can also be used to **test HTTP responses**. Different types of queries and predicates are supported, from XPath and JSONPath on body response, to assert on status code and response headers.

It is well adapted for **REST / JSON APIs**

```
POST https://example.org/api/tests
{
    "id": "4568",
    "evaluate": true
}
HTTP 200
[Asserts]
header "X-Frame-Options" == "SAMEORIGIN"
jsonpath "$.status" == "RUNNING"    # Check the status code
jsonpath "$.tests" count == 25      # Check the number of items
jsonpath "$.id" matches /\d{4}/     # Check the format of the id
```

HTML content

```
GET https://example.org
HTTP 200
[Asserts]
xpath "normalize-space(//head/title)" == "Hello world!"
```

GraphQL

```
POST https://example.org/graphql
```graphql
{
  human(id: "1000") {
    name
    height(unit: FOOT)
  }
}
```
HTTP 200
```

and even **SOAP APIs**

```
POST https://example.org/InStock
Content-Type: application/soap+xml; charset=utf-8
SOAPAction: "http://www.w3.org/2003/05/soap-envelope"
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:m="https://example.org">
  <soap:Header></soap:Header>
  <soap:Body>
    <m:GetStockPrice>
      <m:StockName>GOOG</m:StockName>
    </m:GetStockPrice>
  </soap:Body>
</soap:Envelope>
HTTP 200
```

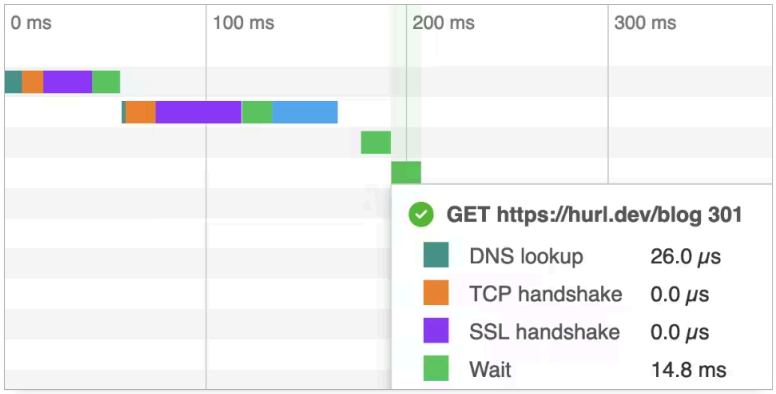
Hurl can also be used to test the **performance** of HTTP endpoints

```
GET https://example.org/api/v1/pets
HTTP 200
[Asserts]
duration < 1000  # Duration in ms
```

And check response bytes

```
GET https://example.org/data.tar.gz
HTTP 200
[Asserts]
sha256 == hex,039058c6f2c0cb492c533b0a4d14ef77cc0f78abccced5287d84a1a2011cfb81;
```

Finally, Hurl is easy to **integrate in CI/CD**, with text, JUnit, TAP and HTML reports



Why Hurl?

Text Format

For both devops and developers

Fast CLI

A command line for local dev and continuous integration

Single Binary

Easy to install, with no runtime required

Powered by curl

Hurl is a lightweight binary written in Rust. Under the hood, Hurl HTTP engine is powered by libcurl, one of the most powerful and reliable file transfer libraries. With its text file format, Hurl adds syntactic sugar to run and test HTTP requests, but it's still the curl that we love: **fast**, **efficient** and **IPv6 / HTTP/3 ready**.

Feedbacks

To support its development, star Hurl on GitHub!

Feedback, suggestion, bugs or improvements are welcome.

```
POST https://hurl.dev/api/feedback
{
  "name": "John Doe",
  "feedback": "Hurl is awesome!"
}
HTTP 200
```

Resources

License

Blog

Tutorial

Documentation

GitHub

Getting Started

Installation

Binaries Installation

Linux

Precompiled binary (depending on libc >=2.35) is available at Hurl latest GitHub release:

```
$ INSTALL_DIR=/tmp
$ VERSION=8.0.1
$ curl --silent --location https://github.com/Orange-
OpenSource/hurl/releases/download/$VERSION/hurl-$VERSION-x86_64-unknown-linux-
gnu.tar.gz | tar xvz -C $INSTALL_DIR
$ export PATH=$INSTALL_DIR/hurl-$VERSION-x86_64-unknown-linux-gnu/bin:$PATH
```

Debian / Ubuntu

For Debian >=12 / Ubuntu 22.04 and 24.04, Hurl can be installed using a binary .deb file provided in each Hurl release.

```
$ VERSION=8.0.1
$ curl --location --remote-name https://github.com/Orange-
OpenSource/hurl/releases/download/$VERSION/hurl_${VERSION}_amd64.deb
$ sudo apt update && sudo apt install ./hurl_${VERSION}_amd64.deb
```

For Ubuntu >=22.04, Hurl can be installed from `ppa:lepapareil/hurl`

```
$ VERSION=8.0.1
$ sudo apt-add-repository -y ppa:lepapareil/hurl
$ sudo apt install hurl="${VERSION}"*
```

Alpine

Hurl is available on `testing` channel.

```
$ apk add --repository http://dl-cdn.alpinelinux.org/alpine/edge/testing hurl
```

Arch Linux / Manjaro

Hurl is available on extra channel.

```
$ pacman -Sy hurl
```

NixOS / Nix

NixOS / Nix package is available on stable channel.

Guix

```
$ guix install hurl
```

macOS

Precompiled binaries for Intel and ARM CPUs are available at Hurl latest GitHub release.

Homebrew

```
$ brew install hurl
```

MacPorts

```
$ sudo port install hurl
```

FreeBSD

```
$ sudo pkg install hurl
```

Windows

Windows requires the Visual C++ Redistributable Package to be installed manually, as this is not included in the installer.

Zip File

Hurl can be installed from a standalone zip file at Hurl latest GitHub release. You will need to update your `PATH` variable.

Installer

An executable installer is also available at Hurl latest GitHub release.

Chocolatey

```
$ choco install hurl
```

Scoop

```
$ scoop install hurl
```

Windows Package Manager

```
$ winget install hurl
```

Cargo

If you're a Rust programmer, Hurl can be installed with cargo.

```
$ cargo install --locked hurl
```

conda-forge

```
$ conda install -c conda-forge hurl
```

Hurl can also be installed with `conda-forge` powered package manager like `pixi`.

Docker

```
$ docker pull ghcr.io/orange-opensource/hurl:latest
```

npm

```
$ npm install --save-dev @orangeopensource/hurl
```

Building From Sources

Hurl sources are available in GitHub.

Build on Linux

Hurl depends on libssl, libcurl and libxml2 native libraries. You will need their development files in your platform.

Debian based distributions

```
$ apt install build-essential pkg-config libssl-dev libcurl4-openssl-dev
libxml2-dev libclang-dev
```

Fedora based distributions

```
$ dnf install pkgconf-pkg-config gcc openssl-devel libxml2-devel clang-devel
```

Red Hat based distributions

```
$ yum install pkg-config gcc openssl-devel libxml2-devel clang-devel
```

Arch based distributions

```
$ pacman -S pkgconf gcc glibc openssl libxml2 clang
```

Alpine based distributions

```
$ apk add curl-dev gcc libxml2-dev musl-dev openssl-dev clang-dev
```

Build on macOS

```
$ xcode-select --install
$ brew install pkg-config openssl
```

Hurl is written in Rust. You should install the latest stable release.

```
$ curl https://sh.rustup.rs -sSf | sh -s -- -y
$ source $HOME/.cargo/env
$ rustc --version
$ cargo --version
```

Then build hurl:

```
$ git clone https://github.com/Orange-OpenSource/hurl
$ cd hurl
$ cargo build --release
$ ./target/release/hurl --version
```

Build on Windows

Please follow the contrib on Windows section.

Manual

Name

hurl - run and test HTTP requests.

Synopsis

hurl OPTIONS [FILES…]

hurl –test OPTIONS [FILES…]

Description

Hurl is a command line tool that runs HTTP requests defined in a simple plain text format.

It can chain requests, capture values and evaluate queries on headers and body response. Hurl is very versatile, it can be used for fetching data and testing HTTP sessions: HTML content, REST / SOAP / GraphQL APIs, or any other XML / JSON based APIs.

```
$ hurl session.hurl
```

If no input files are specified, input is read from stdin.

```
$ echo GET http://httpbin.org/get | hurl
    {
      "args": {},
      "headers": {
        "Accept": "*/*",
        "Accept-Encoding": "gzip",
        "Content-Length": "0",
        "Host": "httpbin.org",
        "User-Agent": "hurl/0.99.10",
        "X-Amzn-Trace-Id": "Root=1-5eedf4c7-520814d64e2f9249ea44e0"
      },
      "origin": "1.2.3.4",
      "url": "http://httpbin.org/get"
    }
```

Hurl can take files as input, or directories. In the latter case, Hurl will search files with `.hurl` extension recursively.

Output goes to stdout by default. To have output go to a file, use the `-o, --output` option:

```
$ hurl -o output input.hurl
```

By default, Hurl executes all HTTP requests and outputs the response body of the last HTTP call.

To have a test oriented output, you can use `--test` option:

```
$ hurl --test *.hurl
```

Hurl File Format

The Hurl file format is fully documented in https://hurl.dev/docs/hurl-file.html

It consists of one or several HTTP requests

```
GET http://example.org/endpoint1
GET http://example.org/endpoint2
```

Capturing values

A value from an HTTP response can be-reused for successive HTTP requests.

A typical example occurs with CSRF tokens.

```
GET https://example.org
HTTP 200
# Capture the CSRF token value from html body.
[Captures]
csrf_token: xpath "normalize-space(//meta[@name='_csrf_token']/@content)"

# Do the login !
POST https://example.org/login?user=toto&password=1234
X-CSRF-TOKEN: {{csrf_token}}
```

More information on captures can be found here https://hurl.dev/docs/capturing-response.html

Asserts

The HTTP response defined in the Hurl file are used to make asserts. Responses are optional.

At the minimum, response includes assert on the HTTP status code.

```
GET http://example.org
HTTP 301
```

It can also include asserts on the response headers

```
GET http://example.org
HTTP 301
Location: http://www.example.org
```

Explicit asserts can be included by combining a query and a predicate

```
GET http://example.org
HTTP 301
[Asserts]
xpath "string(//title)" == "301 Moved"
```

With the addition of asserts, Hurl can be used as a testing tool to run scenarios.

More information on asserts can be found here https://hurl.dev/docs/asserting-response.html

Configuration

Options that exist in curl have exactly the same semantics.

Options specified on the command line are defined for every Hurl file's entry, except if they are tagged as cli-only (can not be defined in the Hurl request `[Options]` entry)

For instance:

```
$ hurl --location foo.hurl
```

will follow redirection for each entry in `foo.hurl`. You can also define an option only for a particular entry with an `[Options]` section. For instance, this Hurl file:

```
GET https://example.org
HTTP 301

GET https://example.org
[Options]
location: true
HTTP 200
```

will follow a redirection only for the second entry.

Hurl looks also for a configuration file at the following places `$XDG_CONFIG_HOME/hurl/config` if `$XDG_CONFIG_HOME` exists, or `$HOME/.config/hurl/config`. The configuration file is a list of options added to each Hurl run.

```
$ cat ~/.config/hurl/config
# Hurl config file
--header x-identifier:quux
--verbose
--delay=1s
--fail-with-body
```

Most of the options can also be defined with environment variables (like `HURL_INSECURE` for `--insecure`). So, in order to configure Hurl, there are four sources from the lowest priority (most easily overridden) to highest (overrides all others):

- Configuration file (ex: `~/.config/hurl/config`)
- Environment variables (ex: `export HURL_INSECURE=true`)
- Command-line options (ex: `hurl --insecure`)
- Options section options (ex: `insecure: true` in file)

All Options

HTTP options

Option	Description
`--aws-sigv4 <PROVIDER1[:PROVIDER2[:REGION[:SERVICE]]]>`	Generate an `Authorization` header with an AWS SigV4 signature. Use `-u, --user` to specify Access Key Id (username) and Secret Key (password). To use temporary session credentials (e.g. for an AWS IAM Role), add the `X-Amz-Security-Token` header containing the session token.
`--cacert <FILE>`	Specifies the certificate file for peer verification. The file may contain multiple CA certificates and must be in PEM format. Normally Hurl is built to use a default file for this, so this option is typically used to alter that default file.
`-E, --cert <CERTIFICATE[:PASSWORD]>`	Client certificate file and password. See also `--key`.
`--compressed`	Request a compressed response using one of the algorithms br, gzip, deflate and automatically decompress the content. Environment variables: HURL_COMPRESSED
`--connect-timeout <SECONDS>`	Maximum time in seconds that you allow Hurl's connection to take. You can specify time units in the connect timeout expression. Set Hurl to use a connect timeout of 20 seconds with `--connect-timeout 20s` or set it to 35,000 milliseconds with `--connect-timeout 35000ms`. No spaces allowed. See also `-m, --max-time`. Environment variables: HURL_CONNECT_TIMEOUT
`--connect-to <HOST1:PORT1:HOST2:PORT2>`	For a request to the given HOST1:PORT1 pair, connect to HOST2:PORT2 instead. This option can be used several times in a

Option	Description
	command line. See also `--resolve`.
`--digest`	Tell Hurl to use HTTP Digest authentication
`-H, --header <NAME:VALUE>`	Add an extra header to include in information sent. Can be used several times in a command. Do not add newlines or carriage returns. Environment variables: HURL_HEADER='name1:value1|name2:value2' (headers are separated by |)
`-0, --http1.0`	Tells Hurl to use HTTP version 1.0 instead of using its internally preferred HTTP version. Environment variables: HURL_HTTP10
`--http1.1`	Tells Hurl to use HTTP version 1.1. Environment variables: HURL_HTTP11
`--http2`	Tells Hurl to use HTTP version 2. For HTTPS, this means Hurl negotiates HTTP/2 in the TLS handshake. Hurl does this by default. For HTTP, this means Hurl attempts to upgrade the request to HTTP/2 using the Upgrade: request header. Environment variables: HURL_HTTP2
`--http2-prior-knowledge`	Tells Hurl to issue a non-TLS HTTP request using HTTP/2 directly without HTTP/1.1 Upgrade. It requires prior knowledge that the server supports HTTP/2 straight away. HTTPS requests still do HTTP/2 the standard way with negotiated protocol versions in the TLS handshake. Environment variables: HURL_HTTP2_PRIOR_KNOWLEDGE
`--http3`	Tells Hurl to try HTTP version 3 to the host in the URL, but fallback to earlier HTTP versions if the HTTP/3 connection establishment fails. HTTP/3 is only available for HTTPS and not for HTTP URLs. Environment variables: HURL_HTTP3

Option	Description
-k, --insecure	This option explicitly allows Hurl to perform “insecure” SSL connections and transfers. Environment variables: HURL_INSECURE
-4, --ipv4	This option tells Hurl to use IPv4 addresses only when resolving host names, and not for example try IPv6. Environment variables: HURL_IPV4
-6, --ipv6	This option tells Hurl to use IPv6 addresses only when resolving host names, and not for example try IPv4. Environment variables: HURL_IPV6
--key <KEY>	Private key file name.
--limit-rate <SPEED>	Specify the maximum transfer rate you want Hurl to use, for both downloads and uploads. This feature is useful if you have a limited pipe and you would like your transfer not to use your entire bandwidth. To make it slower than it otherwise would be. The given speed is measured in bytes/second. Environment variables: HURL_LIMIT_RATE
-L, --location	Follow redirect. To limit the amount of redirects to follow use the --max-redirs option Environment variables: HURL_LOCATION
--location-trusted	Like -L, --location, but allows sending the name + password to all hosts that the site may redirect to. This may or may not introduce a security breach if the site redirects you to a site to which you send your authentication info (which is plaintext in the case of HTTP Basic authentication). Environment variables: HURL_LOCATION_TRUSTED
--max-filesize <BYTES>	Specify the maximum size in bytes of a file to download. If the file requested is larger than this value, the transfer does not start. Environment variables: HURL_MAX_FILESIZE This is a cli-only option.

Option	Description
`--max-redirs <NUM>`	Set maximum number of redirection-followings allowed By default, the limit is set to 50 redirections. Set this option to -1 to make it unlimited. Environment variables: HURL_MAX_REDIRS
`-m, --max-time <SECONDS>`	Maximum time in seconds that you allow a request/response to take. This is the standard timeout. You can specify time units in the maximum time expression. Set Hurl to use a maximum time of 20 seconds with `--max-time 20s` or set it to 35,000 milliseconds with `--max-time 35000ms`. No spaces allowed. See also `--connect-timeout`. Environment variables: HURL_MAX_TIME
`--negotiate`	Tell Hurl to use Negotiate (SPNEGO) authentication.
`--no-cookie-store`	Do not use cookie storage for requests/responses in a file. By default, requests in the same Hurl file share cookie storage, this option deactivates cookie engine. Environment variables: HURL_NO_COOKIE_STORE This is a cli-only option.
`--no-header <NAME>`	Remove a header from information sent. Can be used several times in a command. Environment variables: HURL_NO_HEADER='name1|name2' (names are separated by |)
`--no-proxy <HOST(S)>`	Comma-separated list of hosts which do not use a proxy. Environment variables: no_proxy
`--ntlm`	Tell Hurl to use NTLM authentication
`--path-as-is`	Tell Hurl to not handle sequences of /../ or /./ in the given URL path. Normally Hurl will squash or merge them according to standards but with this option set you tell it not to do that.

Option	Description
--pinnedpubkey <HASHES>	When negotiating a TLS or SSL connection, the server sends a certificate indicating its identity. A public key is extracted from this certificate and if it does not exactly match the public key provided to this option, Hurl aborts the connection before sending or receiving any data.
-x, --proxy <[PROTOCOL://]HOST[:PORT]>	Use the specified proxy. Environment variables: http_proxy https_proxy all_proxy
--proxy-header <HEADER>	Extra header to include in the request when sending to a proxy. These headers are only applied when a proxy is involved; if Hurl connects directly to the target server, they are ignored. Environment variables: HURL_PROXY_HEADER='name1:value1|name2:value2' (headers are separated by |) This is a cli-only option.
--resolve <HOST:PORT:ADDR>	Provide a custom address for a specific host and port pair. Using this, you can make the Hurl requests(s) use a specified address and prevent the otherwise normally resolved address to be used. Consider it a sort of /etc/hosts alternative provided on the command line.
--ssl-no-revoke	(Windows) This option tells Hurl to disable certificate revocation checks. WARNING: this option loosens the SSL security, and by using this flag you ask for exactly that. This is a cli-only option.
--tls-max <VERSION>	Set the maximum allowed TLS version. If the connection is done without TLS, this option has no effect. This includes QUIC-using (HTTP/3) transfers. default Use up to the recommended TLS version. 1.0 Use up to TLSv1.0. 1.1 Use up to TLSv1.1. 1.2 Use up to TLSv1.2. 1.3 Use up to TLSv1.3. Environment variables: HURL_TLS_MAX

Option	Description
`--unix-socket <PATH>`	(HTTP) Connect through this Unix domain socket, instead of using the network.
`-u, --user <USER:PASSWORD>`	Add basic Authentication header to each request. Environment variables: HURL_USER
`-A, --user-agent <NAME>`	Specify the User-Agent string to send to the HTTP server. Environment variables: HURL_USER_AGENT This is a cli-only option.

Output options

Option	Description
`--color`	Colorize standard output and standard error. By default, Hurl outputs a prettified and colorized response. When redirected through pipes, standard streams are not colorized and color can be forced with this option. Environment variables: HURL_COLOR This is a cli-only option.
`--curl <FILE>`	Export each request to a list of curl commands. This is a cli-only option.
`--error-format <FORMAT>`	Control the format of error message (short by default or long). When using long, the response body is logged when there are errors. Environment variables: HURL_ERROR_FORMAT This is a cli-only option.
`--fail-with-body`	Output body response on standard output if there are any errors. By default, Hurl outputs the last body response on standard output only when a run is successful. Environment variables: HURL_FAIL_WITH_BODY

Option	Description
-i, --include	Include the HTTP headers in the output This is a cli-only option.
--json	Output each Hurl file result to JSON. The format is very closed to HAR format. This is a cli-only option.
--no-color	Do not colorize standard output nor standard error. Environment variables: HURL_NO_COLOR NO_COLOR This is a cli-only option.
--no-output	Suppress output. By default, Hurl outputs the body of the last response. Environment variables: HURL_NO_OUTPUT This is a cli-only option.
--no-pretty	Do not prettify response output for supported content type (JSON only for the moment). By default, output is prettified if standard output is a terminal. Environment variables: HURL_NO_PRETTY This is a cli-only option.
-o, --output <FILE>	Write output to FILE instead of stdout. Use '-' for stdout in Options sections.
--pretty	Prettify response output for supported content type (JSON only for the moment). By default, JSON response is prettified if standard output is a terminal, and colorized, see--no-color to format without color. Environment variables: HURL_PRETTY This is a cli-only option.
--progress-bar	Display a progress bar in test mode. The progress bar is displayed only in interactive TTYs. This option forces the progress bar to be displayed even in non-interactive TTYs. This is a cli-only option.

Option	Description
-v, --verbose	Turn on verbose output on standard error stream. Useful for debugging. A line starting with '>' means data sent by Hurl. A line staring with '<' means data received by Hurl. A line starting with '*' means additional info provided by Hurl. If you only want HTTP headers in the output, -i, --include might be the option you're looking for. Environment variables: HURL_VERBOSE
--verbosity <LEVEL>	Set the verbosity level for debug logs on standard error stream (brief, verbose or debug) If you only want HTTP headers in the output, -i, --include might be the option you're looking for. -v, --verbose is an alias for --verbosity verbose --very-verbose is an alias for --verbosity debug Environment variables: HURL_VERBOSITY
--very-verbose	Turn on more verbose output on standard error stream. In contrast to --verbose option, this option outputs the full HTTP body request and response on standard error. In addition, lines starting with '**' are libcurl debug logs. Environment variables: HURL_VERY_VERBOSE

Run options

Option	Description
--continue-on-error	Continue executing requests to the end of the Hurl file even when an assert error occurs. By default, Hurl exits after an assert error in the HTTP response. Note that this option does not affect the behavior with multiple input Hurl files. All the input files are executed independently. The result of one file does not affect the execution of the other Hurl files.

Option	Description
	Environment variables: HURL_CONTINUE_ON_ERROR This is a cli-only option.
`--delay <MILLISECONDS>`	Sets delay before each request (aka sleep). The delay is not applied to requests that have been retried because of `--retry`. See `--retry-interval` to space retried requests. You can specify time units in the delay expression. Set Hurl to use a delay of 2 seconds with `--delay 2s` or set it to 500 milliseconds with `--delay 500ms`. Supported time units: ms, s, m, h. No spaces allowed. Environment variables: HURL_DELAY
`--from-entry <ENTRY_NUMBER>`	Execute Hurl file from ENTRY_NUMBER (starting at 1). This is a cli-only option.
`--jobs <NUM>`	Maximum number of parallel jobs in parallel mode. Default value corresponds (in most cases) to the current amount of CPUs. Set to 1 to disable parallel execution of files. See also `--parallel`. Environment variables: HURL_JOBS This is a cli-only option.
`--no-assert`	Ignore all asserts defined in the Hurl file. Environment variables: HURL_NO_ASSERT This is a cli-only option.
`--no-jsonpath-coercion`	Disable JSONPath result coercion. By default, when JSONPath coercion is enabled, empty JSONPath results are returned as no value and single JSONPath results are returned as a scalar value. With this option, JSONPath results are always returned as arrays. Environment variables: HURL_NO_JSONPATH_COERCION
`--parallel`	Run files in parallel. Each Hurl file is executed in its own worker thread, without sharing

Option	Description
	anything with the other workers. The default run mode is sequential. Parallel execution is by default in `--test` mode. See also `--jobs`. This is a cli-only option.
`--repeat <NUM>`	Repeat the input files sequence NUM times, -1 for infinite loop. Given a.hurl, b.hurl, c.hurl as input, repeat two times will run a.hurl, b.hurl, c.hurl, a.hurl, b.hurl, c.hurl.
`--retry <NUM>`	Maximum number of retries, 0 for no retries, -1 for unlimited retries. Retry happens if any error occurs (asserts, captures, runtimes etc…). Environment variables: HURL_RETRY
`--retry-interval <MILLISECONDS>`	Duration in milliseconds between each retry. Default is 1000 ms. You can specify time units in the retry interval expression. Set Hurl to use a retry interval of 2 seconds with `--retry-interval 2s` or set it to 500 milliseconds with `--retry-interval 500ms`. No spaces allowed. Environment variables: HURL_RETRY_INTERVAL
`--secret <NAME=VALUE>`	Define secret value to be redacted from logs and report. When defined, secrets can be used as variable everywhere variables are used. Environment variables: HURL_SECRET_name This is a cli-only option.
`--secrets-file <FILE>`	Define a secrets file in which you define your secrets Each secret is defined as name=value exactly as with `--secret` option. Note that defining a secret twice produces an error. This is a cli-only option.
`--test`	Activate test mode: with this, the HTTP response is not outputted anymore, progress is reported for each Hurl file tested, and a text summary is displayed when all files have been run. In test mode, files are executed in parallel. To run test in a sequential way use `--jobs 1`.

Option	Description
	See also `--jobs`. Environment variables: HURL_TEST This is a cli-only option.
`--to-entry <ENTRY_NUMBER>`	Execute Hurl file to ENTRY_NUMBER (starting at 1). Ignore the remaining of the file. It is useful for debugging a session. This is a cli-only option.
`--variable <NAME=VALUE>`	Define variable (name/value) to be used in Hurl templates. Environment variables: HURL_VARIABLE_name
`--variables-file <FILE>`	Set properties file in which your define your variables. Each variable is defined as name=value exactly as with `--variable` option. Note that defining a variable twice produces an error. This is a cli-only option.

Report options

Option	Description
`--report-html <DIR>`	Generate HTML report in DIR. If the HTML report already exists, it will be updated with the new test results. This is a cli-only option.
`--report-json <DIR>`	Generate JSON report in DIR. If the JSON report already exists, it will be updated with the new test results. This is a cli-only option.
`--report-junit <FILE>`	Generate JUnit File.

Option	Description
	If the FILE report already exists, it will be updated with the new test results. This is a cli-only option.
–report-tap <FILE>	Generate TAP report. If the FILE report already exists, it will be updated with the new test results. This is a cli-only option.

Other options

Option	Description
–b, –cookie <FILE>	Read cookies from FILE (using the Netscape cookie file format). Combined with –c, ––cookie-jar, you can simulate a cookie storage between successive Hurl runs. This is a cli-only option.
–c, –cookie-jar <FILE>	Write cookies to FILE after running the session. The file will be written using the Netscape cookie file format. Combined with –b, ––cookie, you can simulate a cookie storage between successive Hurl runs. This is a cli-only option.
–file-root <DIR>	Set root directory to import files in Hurl. This is used for files in multipart form data, request body and response output. When it is not explicitly defined, files are relative to the Hurl file's directory. This is a cli-only option.
–glob <GLOB>	Specify input files that match the given glob pattern. Multiple glob flags may be used. This flag supports common Unix glob patterns like *, ? and []. However, to avoid your shell accidentally expanding glob patterns before Hurl handles them, you must use single quotes or double

Option	Description
	quotes around each pattern. This is a cli-only option.
-n, --netrc	Scan the .netrc file in the user's home directory for the username and password. See also --netrc-file and --netrc-optional.
--netrc-file <FILE>	Like --netrc, but provide the path to the netrc file. See also --netrc-optional.
--netrc-optional	Similar to --netrc, but make the .netrc usage optional. See also --netrc-file.
-h, --help	Usage help. This lists all current command line options with a short description.
-V, --version	Prints version information

Exit Codes

Value	Description
0	Success.
1	Failed to parse command-line options.
2	Input file parsing error.
3	Runtime error (such as failure to connect to host).
4	Assert error.

WWW

https://hurl.dev

See Also

Samples

To run a sample, edit a file with the sample content, and run Hurl:

```
$ vi sample.hurl

GET https://example.org

$ hurl sample.hurl
```

By default, Hurl behaves like curl and outputs the last HTTP response's entry. To have a test oriented output, you can use `--test` option:

```
$ hurl --test sample.hurl
```

A particular response can be saved with `[Options]` section:

```
GET https://example.org/cats/123
[Options]
output: cat123.txt    # use - to output to stdout
HTTP 200

GET https://example.org/dogs/567
HTTP 200
```

Finally, Hurl can take files as input, or directories. In the latter case, Hurl will search files with `.hurl` extension recursively.

```
$ hurl --test integration/*.hurl
$ hurl --test .
```

You can check Hurl tests suite for more samples.

Getting Data

A simple GET:

```
GET https://example.org
```

Requests can be chained:

```
GET https://example.org/a
GET https://example.org/b
```

```
HEAD https://example.org/c
GET https://example.org/c
```

Doc

HTTP Headers

A simple GET with headers:

```
GET https://example.org/news
User-Agent: Mozilla/5.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
```

Doc

Query Params

```
GET https://example.org/news
[Query]
order: newest
search: something to search
count: 100
```

Or:

```
GET https://example.org/news?
order=newest&search=something%20to%20search&count=100
```

> With `[Query]` section, params don't need to be URL escaped.

Doc

Basic Authentication

```
GET https://example.org/protected
[BasicAuth]
bob: secret
```

Doc

This is equivalent to construct the request with a Authorization header:

```
# Authorization header value can be computed with `echo -n 'bob:secret' |
base64`
```

```
GET https://example.org/protected
Authorization: Basic Ym9iOnNlY3JldA==
```

Basic authentication section allows per request authentication. If you want to add basic authentication to all the requests of a Hurl file you could use `-u/--user` option:

```
$ hurl --user bob:secret login.hurl
```

`--user` option can also be set per request:

```
GET https://example.org/login
[Options]
user: bob:secret
HTTP 200

GET https://example.org/login
[Options]
user: alice:secret
HTTP 200
```

Passing Data between Requests

Captures can be used to pass data from one request to another:

```
POST https://sample.org/orders
HTTP 201
[Captures]
order_id: jsonpath "$.order.id"

GET https://sample.org/orders/{{order_id}}
HTTP 200
```

Doc

Sending Data

Sending HTML Form Data

```
POST https://example.org/contact
[Form]
default: false
token: {{token}}
email: john.doe@rookie.org
number: 33611223344
```

Doc

Sending Multipart Form Data

```
POST https://example.org/upload
[Multipart]
field1: value1
field2: file,example.txt;
# One can specify the file content type:
field3: file,example.zip; application/zip
```

Doc

Multipart forms can also be sent with a multiline string body:

```
POST https://example.org/upload
Content-Type: multipart/form-data; boundary="boundary"
```
--boundary
Content-Disposition: form-data; name="key1"

value1
--boundary
Content-Disposition: form-data; name="upload1"; filename="data.txt"
Content-Type: text/plain

Hello World!
--boundary
Content-Disposition: form-data; name="upload2"; filename="data.html"
Content-Type: text/html

Hello World!
--boundary--
```
```

In that case, files have to be inlined in the Hurl file.

Doc

Posting a JSON Body

With an inline JSON:

```
POST https://example.org/api/tests
{
    "id": "456",
    "evaluate": true
}
```

Doc

With a local file:

```
POST https://example.org/api/tests
Content-Type: application/json
file,data.json;
```

Doc

Templating a JSON Body

```
PUT https://example.org/api/hits
Content-Type: application/json
{
    "key0": "{{a_string}}",
    "key1": {{a_bool}},
    "key2": {{a_null}},
    "key3": {{a_number}}
}
```

Variables can be initialized via command line:

```
$ hurl --variable a_string=apple \
       --variable a_bool=true \
       --variable a_null=null \
       --variable a_number=42 \
       test.hurl
```

Resulting in a PUT request with the following JSON body:

```
{
    "key0": "apple",
    "key1": true,
    "key2": null,
    "key3": 42
}
```

Doc

Templating a XML Body

Using templates with XML body is not currently supported in Hurl. You can use templates in XML multiline string body with variables to send a variable XML body:

```
POST https://example.org/echo/post/xml
```xml
<?xml version="1.0" encoding="utf-8"?>
<Request>
    <Login>{{login}}</Login>
    <Password>{{password}}</Password>
```
```

````
</Request>
```
````

Doc

Using GraphQL Query

A simple GraphQL query:

````
POST https://example.org/starwars/graphql
```graphql
{
  human(id: "1000") {
    name
    height(unit: FOOT)
  }
}
```
````

A GraphQL query with variables:

````
POST https://example.org/starwars/graphql
```graphql
query Hero($episode: Episode, $withFriends: Boolean!) {
  hero(episode: $episode) {
    name
    friends @include(if: $withFriends) {
      name
    }
  }
}

variables {
  "episode": "JEDI",
  "withFriends": false
}
```
````

GraphQL queries can also use Hurl templates.

Doc

Using Dynamic Datas

Functions like `newUuid` and `newDate` can be used in templates to create dynamic datas:

A file that creates a dynamic email (i.e `0531f78f-7f87-44be-a7f2-969a1c4e6d97@test.com`):

````
POST https://example.org/api/foo
{
  "name": "foo",
````
````

```
  "email": "{{newUuid}}@test.com"
}
```

A file that creates a dynamic query parameter (i.e `2024-12-02T10:35:44.461731Z`):

```
GET https://example.org/api/foo
[Query]
date: {{newDate}}
HTTP 200
```

Doc

Testing Response

Responses are optional, everything after `HTTP` is part of the response asserts.

```
# A request with (almost) no check:
GET https://foo.com

# A status code check:
GET https://foo.com
HTTP 200

# A test on response body
GET https://foo.com
HTTP 200
[Asserts]
jsonpath "$.state" == "running"
```

Testing Status Code

```
GET https://example.org/order/435
HTTP 200
```

Doc

```
GET https://example.org/order/435
# Testing status code is in a 200-300 range
HTTP *
[Asserts]
status >= 200
status < 300
```

Doc

Testing Response Headers

Use implicit response asserts to test header values:

```
GET https://example.org/index.html
HTTP 200
Set-Cookie: theme=light
Set-Cookie: sessionToken=abc123; Expires=Wed, 09 Jun 2021 10:18:14 GMT
```

Doc

Or use explicit response asserts with predicates:

```
GET https://example.org
HTTP 302
[Asserts]
header "Location" contains "www.example.net"
```

Doc

Implicit and explicit asserts can be combined:

```
GET https://example.org/index.html
HTTP 200
Set-Cookie: theme=light
Set-Cookie: sessionToken=abc123; Expires=Wed, 09 Jun 2021 10:18:14 GMT
[Asserts]
header "Location" contains "www.example.net"
```

Testing REST APIs

Asserting JSON body response (node values, collection count etc…) with JSONPath:

```
GET https://example.org/order
screencapability: low
HTTP 200
[Asserts]
jsonpath "$.validated" == true
jsonpath "$.userInfo" isObject
jsonpath "$.userInfo.firstName" == "Franck"
jsonpath "$.userInfo.lastName" == "Herbert"
jsonpath "$.hasDevice" == false
jsonpath "$.links" count == 12
jsonpath "$.state" != null
jsonpath "$.order" matches "^order-\\d{8}$"
jsonpath "$.order" matches /^order-\d{8}$/  # Alternative syntax with regex
literal
jsonpath "$.id" matches /(?i)[a-z]*/        # See syntax for flags
<https://docs.rs/regex/latest/regex/#grouping-and-flags>
jsonpath "$.created" isIsoDate
```

Doc

Testing HTML Response

```
GET https://example.org
HTTP 200
Content-Type: text/html; charset=UTF-8
[Asserts]
xpath "string(/html/head/title)" contains "Example" # Check title
xpath "count(//p)" == 2  # Check the number of p
xpath "//p" count == 2  # Similar assert for p
xpath "boolean(count(//h2))" == false  # Check there is no h2
xpath "//h2" not exists  # Similar assert for h2
xpath "string(//div[1])" matches /Hello.*/
```

Doc

Testing Set-Cookie Attributes

```
GET https://example.org/home
HTTP 200
[Asserts]
cookie "JSESSIONID" == "8400BAFE2F66443613DC38AE3D9D6239"
cookie "JSESSIONID[Value]" == "8400BAFE2F66443613DC38AE3D9D6239"
cookie "JSESSIONID[Expires]" contains "Wed, 13 Jan 2021"
cookie "JSESSIONID[Secure]" exists
cookie "JSESSIONID[HttpOnly]" exists
cookie "JSESSIONID[SameSite]" == "Lax"
```

Doc

Testing Bytes Content

Check the SHA-256 response body hash:

```
GET https://example.org/data.tar.gz
HTTP 200
[Asserts]
sha256 == hex,039058c6f2c0cb492c533b0a4d14ef77cc0f78abccced5287d84a1a2011cfb81;
```

Doc

SSL Certificate

Check the properties of a SSL certificate:

```
GET https://example.org
HTTP 200
[Asserts]
certificate "Subject" == "CN=example.org"
certificate "Issuer" == "C=US, O=Let's Encrypt, CN=R3"
```

```
certificate "Expire-Date" daysAfterNow > 15
certificate "Serial-Number" matches /[\da-f]+/
certificate "Subject-Alt-Name" contains "DNS:example.org"
certificate "Subject-Alt-Name" split "," count == 2
certificate "Value" startsWith "-----BEGIN CERTIFICATE-----"
```

Doc

Checking Full Body

Use implicit body to test an exact JSON body match:

```
GET https://example.org/api/cats/123
HTTP 200
{
  "name" : "Purrsloud",
  "species" : "Cat",
  "favFoods" : ["wet food", "dry food", "<strong>any</strong> food"],
  "birthYear" : 2016,
  "photo" : "https://learnwebcode.github.io/json-example/images/cat-2.jpg"
}
```

Doc

Or an explicit assert file:

```
GET https://example.org/index.html
HTTP 200
[Asserts]
body == file,cat.json;
```

Doc

Implicit asserts supports XML body:

```
GET https://example.org/api/catalog
HTTP 200
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
   <book id="bk101">
      <author>Gambardella, Matthew</author>
      <title>XML Developer's Guide</title>
      <genre>Computer</genre>
      <price>44.95</price>
      <publish_date>2000-10-01</publish_date>
      <description>An in-depth look at creating applications with XML.
</description>
   </book>
</catalog>
```

Doc

Plain text:

```
GET https://example.org/models
HTTP 200
```
Year,Make,Model,Description,Price
1997,Ford,E350,"ac, abs, moon",3000.00
1999,Chevy,"Venture ""Extended Edition""","",4900.00
1999,Chevy,"Venture ""Extended Edition, Very Large""",,5000.00
1996,Jeep,Grand Cherokee,"MUST SELL! air, moon roof, loaded",4799.00
```

Doc

One line:

```
POST https://example.org/helloworld
HTTP 200
`Hello world!`
```

Doc

File:

```
GET https://example.org
HTTP 200
file,data.bin;
```

Doc

Testing Redirections

By default, Hurl doesn't follow redirection so each step of a redirect must be run manually and can be analysed:

```
GET https://example.org/step1
HTTP 301
[Asserts]
header "Location" == "https://example.org/step2"


GET https://example.org/step2
HTTP 301
[Asserts]
header "Location" == "https://example.org/step3"
```
```

```
GET https://example.org/step3
HTTP 200
```

Doc

Using `--location` and `--location-trusted` (either with command line option or per request), Hurl follows redirection and each step of the redirection can be checked.

```
GET https://example.org/step1
[Options]
location: true
HTTP 200
[Asserts]
redirects count == 2
redirects nth 0 location == "https://example.org/step2"
redirects nth 1 location == "https://example.org/step3"
```

```
GET https://example.org/step1
[Options]
location-trusted: true
HTTP 200
[Asserts]
redirects last location == "https://example.org/step2"
```

Doc

Debug Tips

Verbose Mode

To get more info on a given request/response, use `[Options]` section:

```
GET https://example.org
HTTP 200

GET https://example.org/api/cats/123
[Options]
very-verbose: true
HTTP 200
```

`--verbose` and `--very-verbose` can be also used globally as command line options.

Doc

Error Format

```
$ hurl --test --error-format long *.hurl
```

Doc

Output Response Body

Use `--output` on a specific request to get the response body (- can be used as standard output):

```
GET https://foo.com/failure
[Options]
# use - to output on standard output, foo.bin to save on disk
output: -
HTTP 200

GET https://foo.com/success
HTTP 200
```

Doc

Export curl Commands

```
$ hurl ---curl /tmp/curl.txt *.hurl
```

Doc

Using Proxy

Use `--proxy` on a specific request or globally as command line option:

```
GET https://foo.com/a
HTTP 200

GET https://foo.com/b
[Options]
proxy: localhost:8888
HTTP 200

GET https://foo.com/c
HTTP 200
```

Reports

HTML Report

```
$ hurl --test --report-html build/report/ *.hurl
```

Doc

JSON Report

```
$ hurl --test --report-json build/report/ *.hurl
```

Doc

JUnit Report

```
$ hurl --test --report-junit build/report.xml *.hurl
```

Doc

TAP Report

```
$ hurl --test --report-tap build/report.txt *.hurl
```

Doc

JSON Output

A structured output of running Hurl files can be obtained with `--json` option. Each file will produce a JSON export of the run.

```
$ hurl --json *.hurl
```

Others

HTTP Version

Testing HTTP version (HTTP/1.0, HTTP/1.1, HTTP/2 or HTTP/3) can be done using implicit asserts:

```
GET https://foo.com
HTTP/3 200

GET https://bar.com
HTTP/2 200
```

Doc

Or explicit:

```
GET https://foo.com
HTTP 200
[Asserts]
version == "3"

GET https://bar.com
HTTP 200
```

```
[Asserts]
version == "2"
version toFloat > 1.1
```

Doc

IP Address

Testing the IP address of the response, as a string. This string may be IPv6 address:

```
GET https://foo.com
HTTP 200
[Asserts]
ip == "2001:0db8:85a3:0000:0000:8a2e:0370:733"
ip startsWith "2001"
ip isIpv6
```

Polling and Retry

Retry request on any errors (asserts, captures, status code, runtime etc…):

```
# Create a new job
POST https://api.example.org/jobs
HTTP 201
[Captures]
job_id: jsonpath "$.id"
[Asserts]
jsonpath "$.state" == "RUNNING"


# Pull job status until it is completed
GET https://api.example.org/jobs/{{job_id}}
[Options]
retry: 10   # maximum number of retry, -1 for unlimited
retry-interval: 500ms
HTTP 200
[Asserts]
jsonpath "$.state" == "COMPLETED"
```

Doc

Delaying Requests

Add delay for every request, or a particular request:

```
# Delaying this request by 5 seconds (aka sleep)
GET https://example.org/turtle
[Options]
delay: 5s
HTTP 200
```

```
# No delay!
GET https://example.org/turtle
HTTP 200
```

Doc

Skipping Requests

```
# a, c, d are run, b is skipped
GET https://example.org/a

GET https://example.org/b
[Options]
skip: true

GET https://example.org/c

GET https://example.org/d
```

Doc

Testing Endpoint Performance

```
GET https://sample.org/helloworld
HTTP *
[Asserts]
duration < 1000   # Check that response time is less than one second
```

Doc

Using SOAP APIs

```
POST https://example.org/InStock
Content-Type: application/soap+xml; charset=utf-8
SOAPAction: "http://www.w3.org/2003/05/soap-envelope"
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:m="https://example.org">
  <soap:Header></soap:Header>
  <soap:Body>
    <m:GetStockPrice>
      <m:StockName>GOOG</m:StockName>
    </m:GetStockPrice>
  </soap:Body>
</soap:Envelope>
HTTP 200
```

Doc

Capturing and Using a CSRF Token

```
GET https://example.org
HTTP 200
[Captures]
csrf_token: xpath "string(//meta[@name='_csrf_token']/@content)"


POST https://example.org/login?user=toto&password=1234
X-CSRF-TOKEN: {{csrf_token}}
HTTP 302
```

Doc

Redacting Secrets

Using command-line for known values:

```
$ hurl --secret token=1234 file.hurl
```

```
POST https://example.org
X-Token: {{token}}
{
  "name": "Alice",
  "value": 100
}
HTTP 200
```

Doc

Using `redact` for dynamic values:

```
# Get an authorization token:
GET https://example.org/token
HTTP 200
[Captures]
token: header "X-Token" redact

# Send an authorized request:
POST https://example.org
X-Token: {{token}}
{
  "name": "Alice",
  "value": 100
}
HTTP 200
```

Doc

Checking Byte Order Mark (BOM) in Response Body

```
GET https://example.org/data.bin
HTTP 200
[Asserts]
bytes startsWith hex,efbbbf;
```

Doc

AWS Signature Version 4 Requests

Generate signed API requests with AWS Signature Version 4, as used by several cloud providers.

```
POST https://sts.eu-central-1.amazonaws.com/
[Options]
aws-sigv4: aws:amz:eu-central-1:sts
[Form]
Action: GetCallerIdentity
Version: 2011-06-15
```

The Access Key is given per `--user`, either with command line option or within the `[Options]` section:

```
POST https://sts.eu-central-1.amazonaws.com/
[Options]
aws-sigv4: aws:amz:eu-central-1:sts
user: bob=secret
[Form]
Action: GetCallerIdentity
Version: 2011-06-15
```

Doc

Using curl Options

curl options (for instance `--resolve` or `--connect-to`) can be used as CLI argument. In this case, they're applicable to each request of an Hurl file.

```
$ hurl --resolve foo.com:8000:127.0.0.1 foo.hurl
```

Use `[Options]` section to configure a specific request:

```
GET http://bar.com
HTTP 200


GET http://foo.com:8000/resolve
[Options]
```

```
resolve: foo.com:8000:127.0.0.1
HTTP 200
`Hello World!`
```

Doc

Running Tests

Use –test Option

Hurl is run by default as an HTTP client, returning the body of the last HTTP response.

```
$ hurl hello.hurl
Hello World!
```

When multiple input files are provided, Hurl outputs the body of the last HTTP response of each file.

```
$ hurl hello.hurl assert_json.hurl
Hello World![
  { "id": 1, "name": "Bob"},
  { "id": 2, "name": "Bill"}
]
```

For testing, we are only interested in the asserts results, we don't need the HTTP body response. To use Hurl as a test tool with an adapted output, you can use `--test` option:

```
$ hurl --test hello.hurl assert_json.hurl
hello.hurl: Success (6 request(s) in 245 ms)
assert_json.hurl: Success (8 request(s) in 308 ms)
--------------------------------------------------------------------------------
-
Executed files:    2
Executed requests: 10 (17.82/s)
Succeeded files:   2 (100.0%)
Failed files:      0 (0.0%)
Duration:          561 ms
```

Or, in case of errors:

```
$ hurl --test hello.hurl error_assert_status.hurl
hello.hurl: Success (4 request(s) in 5 ms)
error: Assert status code
  --> error_assert_status.hurl:9:6
   |
   | GET http://localhost:8000/not_found
   | ...
 9 | HTTP 200
   |      ^^^ actual value is <404>
   |

error_assert_status.hurl: Failure (1 request(s) in 2 ms)
--------------------------------------------------------------------------------
-
```

```
Executed files:    2
Executed requests: 5 (500.0/s)
Succeeded files:   1 (50.0%)
Failed files:      1 (50.0%)
Duration:          10 ms
```

> With or without `--test`, all asserts are always executed. `--test` adds a run recap and disables the output of the last response. To ignore asserts execution, you can use `--ignore-asserts`.

In test mode, files are executed in parallel to speed-ud the execution. If a sequential run is needed, you can use `--jobs 1` option to execute tests one by one.

```
$ hurl --test --jobs 1 *.hurl
```

`--repeat` option can be used to repeat run files and do performance check. For instance, this call will run 1000 tests in parallel:

```
$ hurl --test --repeat 1000 stress.hurl
```

Selecting Tests

Hurl can take multiple files into inputs:

```
$ hurl --test test/integration/a.hurl test/integration/b.hurl
test/integration/c.hurl
```

```
$ hurl --test test/integration/*.hurl
```

Or you can simply give a directory and Hurl will find files with `.hurl` extension recursively:

```
$ hurl --test test/integration/
```

Finally, you can use `--glob` option to test files that match a given pattern:

```
$ hurl --test --glob "test/integration/**/*.hurl"
```

Debugging

Debug Logs

If you need more error context, you can use `--error-format long` option to print HTTP bodies for failed asserts:

```
$ hurl --test --error-format long hello.hurl error_assert_status.hurl
hello.hurl: Success (4 request(s) in 6 ms)
HTTP/1.1 404
Server: Werkzeug/3.0.3 Python/3.12.4
Date: Wed, 10 Jul 2024 15:42:41 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 207
Server: Flask Server
Connection: close

<!doctype html>
<html lang=en>
<title>404 Not Found</title>
<h1>Not Found</h1>
<p>The requested URL was not found on the server. If you entered the URL
manually please check your spelling and try again.</p>


error: Assert status code
  --> error_assert_status.hurl:9:6
   |
   | GET http://localhost:8000/not_found
   | ...
 9 | HTTP 200
   |      ^^^ actual value is <404>
   |

error_assert_status.hurl: Failure (1 request(s) in 2 ms)
--------------------------------------------------------------------------------
-
Executed files:    2
Executed requests: 5 (454.5/s)
Succeeded files:   1 (50.0%)
Failed files:      1 (50.0%)
Duration:          11 ms
```

Individual requests can be modified with [`[Options]` section](#file-format-request-options) to turn on logs for a particular request, using `verbose` and `very-verbose` option.

With this Hurl file:

```
GET https://foo.com
HTTP 200

GET https://bar.com
[Options]
very-verbose: true
HTTP 200

GET https://baz.com
HTTP 200
```

Running `hurl --test .` will output debug logs for the request to `https://bar.com`.

`--verbose` / `--very-verbose` can also be enabled globally, for every requests of every tested files:

```
$ hurl --test --very-verbose .
```

HTTP Responses

In test mode, HTTP responses are not displayed. One way to get HTTP responses even in test mode is to use `--output` option of `[Options]` section: `--output file` allows to save a particular response to a file, while `--output -` allows to redirect HTTP responses to standard output.

```
GET http://foo.com
HTTP 200

GET https://bar.com
[Options]
output: -
HTTP 200
```

```
$ hurl --test .
<html><head><meta http-equiv="content-type" content="text/html;charset=utf-8">
<title>301 Moved</TITLE></head><body>
<h1>301 Moved</h1>
The document has moved
<a HREF="https://www.bar.com/">here</a>.
</body></html>
/tmp/test.hurl: Success (2 request(s) in 184 ms)
--------------------------------------------------------------------------------
-
Executed files:    1
Executed requests: 2 (10.7/s)
Succeeded files:   1 (100.0%)
Failed files:      0 (0.0%)
Duration:          187 ms
```

Stress and Performance Tests

Hurl can be used to perform stress tests:

- with query duration:

```
GET https://example.org/foo
HTTP 200
[Asserts]
duration < 1200
```

- response metrics with `--very-verbose` or `--json` to get structured timing data:

```
$ hurl --json foo.hurl | jq
...
          "timings": {
            "app_connect": 0,
            "begin_call": "2025-10-06T07:00:57.127794Z",
            "connect": 120915,
            "end_call": "2025-10-06T07:00:57.280724Z",
            "name_lookup": 92987,
            "pre_transfer": 121014,
            "start_transfer": 152721,
            "total": 152807
          }
...
```

In performance uses-cases, it's important to know how Hurl is working to get the best request per second load. Each Hurl file is processed by its own libcurl instance (roughly speaking a living HTTP connection). Let's say we want to stress our application with 100,000 HTTP requests and see how it's behaving. The simplest idea would be to make a single Hurl file repeating 100,000 requests:

```
GET https://my-website/health
[Options]
repeat: 100000
```

and run it:

```
$ hurl --test perf.hurl
```

With a single Hurl file, we won't benefit from any parallel runs (files are run in parallel, requests *within* a file are run sequentially). To benefit from parallel run, we can use `--repeat` as a CLI argument to repeat file execution. It's as if we've written 100,000 identical Hurl files and run them:

```
GET https://my-website/health
```

and run it:

```
$ hurl --test --repeat 100000 perf.hurl
```

In this case, we're running 100,000 Hurl files in parallel. Now, we have another problem: as each file is basically a HTTP connection, we can run out of TCP port, a phenomenon known as ephemeral ports exhaustion. So, to recap:

- running 100,000 requests in a single file won't benefit from parallel run
- running 100,000 one request files can lead to TCP port resources issues

The solution is to combine the two approaches: running 10 files of 10,000 HTTP requests. With 10 files of 10,000 requests, we're going to execute those files in parallel in their own worker. Each worker will be working sequentially, maximizing the usage and not running into TCP ports

exhaustion. We can force jobs count to be exactly 10 so all workers start at the same time and no one is waiting to get a job done:

```
GET https://my-website/health
[Options]
repeat: 10000
```

```
$ hurl --test --repeat 10 --jobs 10 perf.hurl
```

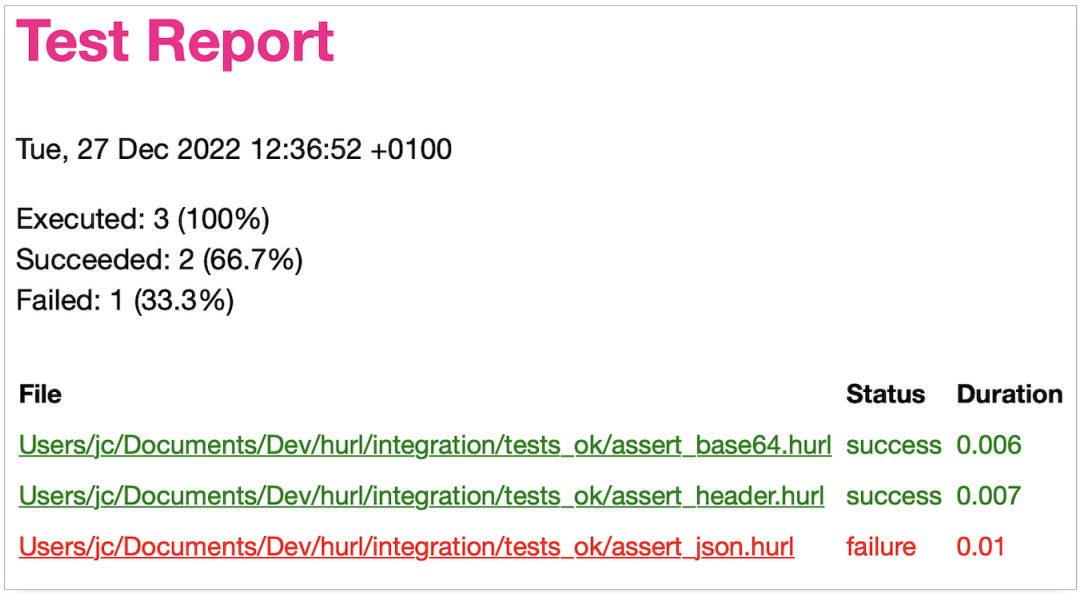
Generating Report

In the different reports, files are always referenced in the input order (which, as tests are executed in parallel, can be different from the execution order).

HTML Report

Hurl can generate an HTML report by using the `--report-html DIR` option.

If the HTML report already exists, the test results will be appended to it.



Test Report

Tue, 27 Dec 2022 12:36:52 +0100

Executed: 3 (100%)
Succeeded: 2 (66.7%)
Failed: 1 (33.3%)

File	Status	Duration
Users/jc/Documents/Dev/hurl/integration/tests_ok/assert_base64.hurl	success	0.006
Users/jc/Documents/Dev/hurl/integration/tests_ok/assert_header.hurl	success	0.007
Users/jc/Documents/Dev/hurl/integration/tests_ok/assert_json.hurl	failure	0.01

The input Hurl files (HTML version) are also included and are easily accessed from the main page.

```
1  GET http://localhost:8000/assert-json
2  HTTP/1.0 200
3  [Asserts]
4  jsonpath "$.count" equals 5
5  jsonpath "$.count" == 5
6  jsonpath "$.count" equals 5.0
7  jsonpath "$.count" notEquals 4
8  jsonpath "$.count" != 4
9  jsonpath "$.count" greaterThan 1
10 jsonpath "$.count" greaterThan 1.0
11 jsonpath "$.count" >= 1.0
```

JSON Report

A JSON report can be produced by using the `--report-json DIR`. The report directory will contain a `report.json` file, including each test file executed with `--json` option and a reference to each HTTP response of the run dumped to disk.

If the JSON report already exists, it will be updated with the new test results.

JUnit Report

A JUnit report can be produced by using the `--report-junit FILE` option.

If the JUnit report already exists, it will be updated with the new test results.

TAP Report

A TAP report (Test Anything Protocol) can be produced by using the `--report-tap FILE` option.

If the TAP report already exists, it will be updated with the new test results.

Use Variables in Tests

To use variables in your tests, you can:

- use `--variable` option
- use `--variables-file` option
- define environment variables, for instance `HURL_VARIABLE_foo=bar`
- define variables in an `[Options]` section(#file-format-request-options)
- define variables in Hurl configuration file

You will find a detailed description in the Injecting Variables section of the docs.

Frequently Asked Questions

General

Why “Hurl”?

The name Hurl is a tribute to the awesome curl, with a focus on the HTTP protocol. While it may have an informal meaning not particularly elegant, other eminent tools have set a precedent in naming.

Yet Another Tool, I already use X

We think that Hurl has some advantages compared to similar tools.

Hurl is foremost a command line tool and should be easy to use on a local computer, or in a CI/CD pipeline. Some tools in the same space as Hurl (Postman for instance), are GUI oriented, and we find it less attractive than CLI. As a command line tool, Hurl can be used to get HTTP data (like curl), but also as a test tool for HTTP sessions, or even as documentation.

Having a text based file format is another advantage. The Hurl format is simple, focused on the HTTP domain, can serve as documentation and can be read or written by non-technical people.

For instance putting JSON data with Hurl can be done with this simple file:

```
PUT http://localhost:3000/api/login
{
  "username": "xyz",
  "password": "xyz"
}
```

With curl:

```
curl --header "Content-Type: application/json" \
     --request PUT \
     --data '{"username": "xyz","password": "xyz"}' \
     http://localhost:3000/api/login
```

Karate, a tool combining API test automation, mocking, performance-testing, has similar features but offers also much more at a cost of an increased complexity.

Comparing Karate file format:

```
Scenario: create and retrieve a cat

Given url 'http://myhost.com/v1/cats'
And request { name: 'Billie' }
When method post
```

```
Then status 201
And match response == { id: '#notnull', name: 'Billie }

Given path response.id
When method get
Then status 200
```

And Hurl:

```
# Scenario: create and retrieve a cat

POST http://myhost.com/v1/cats
{ "name": "Billie" }
HTTP 201
[Captures]
cat_id: jsonpath "$.id"
[Asserts]
jsonpath "$.name" == "Billie"

GET http://myshost.com/v1/cats/{{cat_id}}
HTTP 200
```

A key point of Hurl is to work on the HTTP domain. In particular, there is no JavaScript runtime, Hurl works on the raw HTTP requests/responses, and not on a DOM managed by a HTML engine. For security, this can be seen as a feature: let's say you want to test backend validation, you want to be able to bypass the browser or javascript validations and directly test a backend endpoint.

Finally, with no headless browser and working on the raw HTTP data, Hurl is also really reliable with a very small probability of false positives. Integration tests with tools like Selenium can, in this regard, be challenging to maintain.

Just use what is convenient for you. In our case, it's Hurl!

Hurl is build on top of libcurl, but what is added?

Hurl has two main functionalities on top of curl:

1. Chain several requests:

   With its captures, it enables to inject data received from a response into following requests. CSRF tokens are typical examples in a standard web session.

2. Test HTTP responses:

   With its asserts, responses can be easily tested.

Hurl benefits from the features of the `libcurl` against it is linked. You can check `libcurl` version with `hurl --version`.

For instance on macOS:

```
$ hurl --version
hurl 2.0.0 libcurl/7.79.1 (SecureTransport) LibreSSL/3.3.6 zlib/1.2.11
nghttp2/1.45.1
Features (libcurl):  alt-svc AsynchDNS HSTS HTTP2 IPv6 Largefile libz NTLM
NTLM_WB SPNEGO SSL UnixSockets
Features (built-in): brotli
```

You can also check which `libcurl` is used.

On macOS:

```
$ which hurl
/opt/homebrew/bin/hurl
$ otool -L /opt/homebrew/bin/hurl:
    /usr/lib/libxml2.2.dylib (compatibility version 10.0.0, current version
10.9.0)

/System/Library/Frameworks/CoreFoundation.framework/Versions/A/CoreFoundation
(compatibility version 150.0.0, current version 1858.112.0)
    /usr/lib/libcurl.4.dylib (compatibility version 7.0.0, current version
9.0.0)
    /usr/lib/libiconv.2.dylib (compatibility version 7.0.0, current version
7.0.0)
    /usr/lib/libSystem.B.dylib (compatibility version 1.0.0, current version
1311.100.3)
```

On Linux:

```
$ which hurl
/root/.cargo/bin/hurl
$ ldd /root/.cargo/bin/hurl
ldd /root/.cargo/bin/hurl
    linux-vdso.so.1 (0x0000ffff8656a000)
    libxml2.so.2 => /usr/lib/aarch64-linux-gnu/libxml2.so.2
(0x0000ffff85fe8000)
    libcurl.so.4 => /usr/lib/aarch64-linux-gnu/libcurl.so.4
(0x0000ffff85f45000)
    libgcc_s.so.1 => /lib/aarch64-linux-gnu/libgcc_s.so.1 (0x0000ffff85f21000)
    ...
    libkeyutils.so.1 => /lib/aarch64-linux-gnu/libkeyutils.so.1
(0x0000ffff82ed5000)
    libffi.so.7 => /usr/lib/aarch64-linux-gnu/libffi.so.7 (0x0000ffff82ebc000)
```

Note that some Hurl features are dependent on `libcurl` capacities: for instance, if your `libcurl` doesn't support HTTP/2 Hurl won't be able to send HTTP/2 request.

Why shouldn't I use Hurl?

If you need a GUI. Currently, Hurl does not offer a GUI version (like Postman). While we think that it can be useful, we prefer to focus for the time-being on the core, keeping something simple and fast.

Contributions to build a GUI are welcome.

I have a large numbers of tests, how to run just specific tests?

By convention, you can organize Hurl files into different folders or prefix them.

For example, you can split your tests into two folders critical and additional.

```
critical/test1.hurl
critical/test2.hurl
additional/test1.hurl
additional/test2.hurl
```

You can simply run your critical tests with

```
$ hurl --test critical/*.hurl
```

How can I use my Hurl files outside Hurl?

Hurl file can be exported to a JSON file with `hurlfmt`. This JSON file can then be easily parsed for converting a different format, getting ad-hoc information,…

For example, the Hurl file

```
GET https://example.org/api/users/1
User-Agent: Custom
HTTP 200
[Asserts]
jsonpath "$.name" == "Bob"
```

will be converted to JSON with the following command:

```
$ hurlfmt test.hurl --out json | jq
{
  "entries": [
    {
      "request": {
        "method": "GET",
        "url": "https://example.org/api/users/1",
        "headers": [
          {
            "name": "User-Agent",
            "value": "Custom"
          }
        ]
      },
      "response": {
        "version": "HTTP",
        "status": 200,
        "asserts": [
```

```
          {
            "query": {
              "type": "jsonpath",
              "expr": "$.name"
            },
            "predicate": {
              "type": "==",
              "value": "Bob"
            }
          }
        ]
      }
    }
  ]
}
```

Can I do calculation within a Hurl file?

Currently, the templating is very simple, only accessing variables. Calculations can be done beforehand, before running the Hurl File.

For example, with date calculations, variables `now` and `tomorrow` can be used as param or expected value.

```
$ TODAY=$(date '+%y%m%d')
$ TOMORROW=$(date '+%y%m%d' -d"+1days")
$ hurl --variable "today=$TODAY" --variable "tomorrow=$TOMORROW" test.hurl
```

You can also use environment variables that begins with `HURL_VARIABLE_` to inject data in an Hurl file. For instance, to inject `today` and `tomorrow` variables:

```
$ export HURL_VARIABLE_today=$(date '+%y%m%d')
$ export HURL_VARIABLE_tomorrow=$(date '+%y%m%d' -d"+1days")
$ hurl test.hurl
```

You can also use filters to process HTTP responses in asserts and captures.

macOS

How can I use a custom libcurl (from Homebrew by instance)?

No matter how you've installed Hurl (using the precompiled binary for macOS or with Homebrew) Hurl is linked against the built-in system libcurl. If you want to use another libcurl (for instance, if you've installed curl with Homebrew and want Hurl to use Homebrew's libcurl), you can patch Hurl with the following command:

```
$ sudo install_name_tool -change /usr/lib/libcurl.4.dylib
PATH_TO_CUSTOM_LIBCURL PATH_TO_HURL_BIN
```

For instance:

```
# /usr/local/opt/curl/lib/libcurl.4.dylib is installed by `brew install curl`
$ sudo install_name_tool -change /usr/lib/libcurl.4.dylib
/usr/local/opt/curl/lib/libcurl.4.dylib /usr/local/bin/hurl
```

File Format

Hurl File

Character Encoding

Hurl file should be encoded in UTF-8, without a byte order mark at the beginning (while Hurl ignores the presence of a byte order mark rather than treating it as an error)

File Extension

Hurl file extension is `.hurl`

Comments

Comments begin with # and continue until the end of line. Hurl file can serve as a documentation for HTTP based workflows so it can be useful to be very descriptive.

```
# A very simple Hurl file
# with tasty comments...
GET https://www.sample.net
x-app: MY_APP  # Add a dummy header
HTTP 302       # Check that we have a redirection
[Asserts]
header "Location" exists
header "Location" contains "login"  # Check that we are redirected to the login
page
```

Special Characters in Strings

String can include the following special characters:

- The escaped special characters " (double quotation mark), \ (backslash), backspace), form feed), (line feed), carriage return), and horizontal tab)
- An arbitrary Unicode scalar value, written as , where n is a 1–8 digit hexadecimal number

```
GET https://example.org/api
HTTP 200
# The following assert are equivalent:
[Asserts]
jsonpath "$.slideshow.title" == "A beautiful ✈!"
jsonpath "$.slideshow.title" == "A beautiful \u{2708}!"
```

In some case, (in headers value, etc..), you will also need to escape # to distinguish it from a comment. In the following example:

```
GET https://example.org/api
x-token: BEEF \#STEAK # Some comment
HTTP 200
```

We're sending a header `x-token` with value `BEEF #STEAK`

Entry

Definition

A Hurl file is a list of entries, each entry being a mandatory request, optionally followed by a response.

Responses are not mandatory, a Hurl file consisting only of requests is perfectly valid. To sum up, responses can be used to capture values to perform subsequent requests, or add asserts to HTTP responses.

Example

```
# First, test home title.
GET https://acmecorp.net
HTTP 200
[Asserts]
xpath "normalize-space(//head/title)" == "Hello world!"

# Get some news, response description is optional
GET https://acmecorp.net/news

# Do a POST request without CSRF token and check
# that status code is Forbidden 403
POST https://acmecorp.net/contact
[Form]
default: false
email: john.doe@rookie.org
number: 33611223344
HTTP 403
```

Description

Options

Options specified on the command line apply to every entry in an Hurl file. For instance, with `--location` option, every entry of a given file will follow redirection:

```
$ hurl --location foo.hurl
```

You can use an `[Options]` section(#file-format-request-options) to set option only for a specified request. For instance, in this Hurl file, the second entry will follow location (so we can test the status code to be 200 instead of 301).

```
GET https://google.fr
HTTP 301

GET https://google.fr
[Options]
location: true
HTTP 200

GET https://google.fr
HTTP 301
```

You can use the `[Options](#getting-started-manual-options)` section to log a specific entry:

```
# ... previous entries

GET https://api.example.org
[Options]
very-verbose: true
HTTP 200

# ... next entries
```

Cookie storage

By default, requests in the same Hurl file share cookie storage, enabling session-based scenario. The shared cookie store can be disabled with [`--no-cookie-store`] option.

Redirects

By default, Hurl doesn't follow redirection. To effectively run a redirection, entries should describe each step of the redirection, allowing insertion of asserts in each response.

```
# First entry, test the redirection (status code and 'Location' header)
GET https://example.org
HTTP 301
Location: https://www.example.org

# Second entry, the 200 OK response
GET https://www.example.org
HTTP 200
```

Alternatively, one can use `--location` / `--location-trusted` options to force redirection to be followed. In this case, asserts are executed on the last received response. Optionally, the number of redirections can be limited with `--max-redirs`.

```
# Running hurl --location foo.hurl
GET https://example.org
HTTP 200
```

Finally, you can force redirection on a particular request with an [Options] section(#file-format-request-options) and the --location / --location-trusted options:

```
GET https://example.org
[Options]
location-trusted: true
HTTP 200
```

Redirections can be tested either by:

- running and asserting each step of redirection:

```
GET https://example.org/step1
HTTP 301
[Asserts]
header "Location" == "https://example.org/step2"


GET https://example.org/step2
HTTP 301
[Asserts]
header "Location" == "https://example.org/step3"


GET https://example.org/step3
HTTP 200
```

- using --location / --location-trusted, testing each step with redirects query:

```
GET https://example.org/step1
[Options]
location: true
HTTP 200
[Asserts]
redirects count == 2
redirects nth 0 location == "https://example.org/step2"
redirects nth 1 location == "https://example.org/step3"
```

url query can also be used to get the final effective URL:

```
GET https://example.org/step1
[Options]
location: true
HTTP 200
[Asserts]
url == "https://example.org/step3"
```

Retry

Every entry can be retried upon asserts, captures or runtime errors. Retries allow polling scenarios and effective runs under flaky conditions. Asserts can be explicit (with an [Asserts] section(#file-format-response-asserts)), or implicit (like headers or status code).

Retries can be set globally for every request (see --retry and --retry-interval), or activated on a particular request with an [Options] section(#file-format-request-options).

For example, in this Hurl file, first we create a new job then we poll the new job until it's completed:

```
# Create a new job
POST http://api.example.org/jobs
HTTP 201
[Captures]
job_id: jsonpath "$.id"
[Asserts]
jsonpath "$.state" == "RUNNING"


# Pull job status until it is completed
GET http://api.example.org/jobs/{{job_id}}
[Options]
retry: 10   # maximum number of retry, -1 for unlimited
retry-interval: 300ms
HTTP 200
[Asserts]
jsonpath "$.state" == "COMPLETED"
```

Control flow

In [Options](#getting-started-manual-options) section, skip and repeat can be used to control flow of execution:

- skip: true/false skip this request and execute the next one unconditionally,
- repeat: N loop the request N times. If there are assert or runtime errors, the requests execution is stopped.

```
# This request will be played exactly 3 times
GET https://example.org/foo
[Options]
repeat: 3
HTTP 200

# This request is skipped
GET https://example.org/foo
[Options]
skip: true
HTTP 200
```

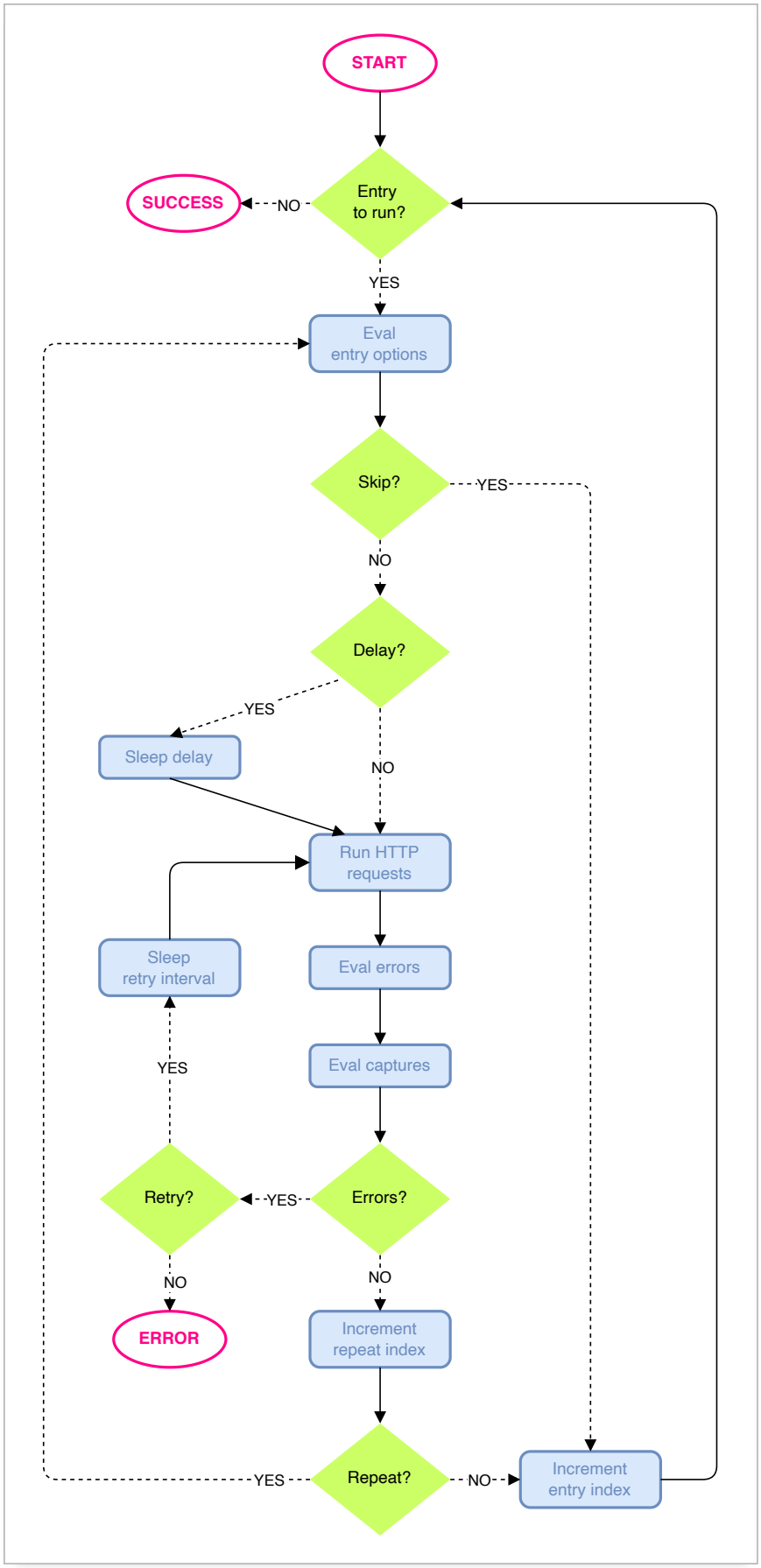
Additionally, a delay can be inserted between requests, to add a delay before execution of a request (aka sleep).

```
# A 5 seconds delayed request
GET https://example.org/foo
[Options]
delay: 5s
HTTP 200
```

delay and repeat can also be used globally as command line options:

```
$ hurl --delay 500ms --repeat 3 foo.hurl
```

For complete reference, below is a diagram for the executed entries.


START
Entry to run?
NO
SUCCESS
YES
Eval entry options
Skip?
YES
NO
Delay?
YES
Sleep delay
NO
Run HTTP requests
Sleep retry interval
Eval errors
YES
Eval captures
Retry?
YES
Errors?
NO
ERROR
NO
Increment repeat index
Repeat?
YES
NO
Increment entry index

Request

Definition

Request describes an HTTP request: a mandatory method and URL, followed by optional headers.

Then, options, query parameters, form parameters, multipart form data, cookies, and basic authentication can be used to configure the HTTP request.

Finally, an optional body can be used to configure the HTTP request body.

Example

```
GET https://example.org/api/dogs?id=4567
User-Agent: My User Agent
Content-Type: application/json
[BasicAuth]
alice: secret
```

Structure



Headers, if present, follow directly after the method and URL. This allows Hurl format to 'look like' the real HTTP format. Contrary to HTTP headers, other parameters are defined in sections (`[Cookies]`, `[Query]`, `[Form]` etc...) These sections are not ordered and can be mixed in any way:

```
GET https://example.org/api/dogs
User-Agent: My User Agent
[Query]
id: 4567
order: newest
[BasicAuth]
alice: secret
```

```
GET https://example.org/api/dogs
User-Agent: My User Agent
[BasicAuth]
alice: secret
[Query]
id: 4567
order: newest
```

The last optional part of a request configuration is the request body. Request body must be the last parameter of a request (after headers and request sections). Like headers, body have no explicit marker:

```
POST https://example.org/api/dogs?id=4567
User-Agent: My User Agent
{
 "name": "Ralphy"
}
```

Description

Method

Mandatory HTTP request method, usually one of `GET`, `HEAD`, `POST`, `PUT`, `DELETE`, `CONNECT`, `OPTIONS`, `TRACE` and `PATCH`.

> Other methods can be used like `QUERY` with the constraint of using only uppercase chars.

URL

Mandatory HTTP request URL.

URL can contain query parameters, even if using a query parameters section is preferred.

```
# A request with URL containing query parameters.
GET https://example.org/forum/questions/?search=Install%20Linux&order=newest
```

```
# A request with query parameters section, equivalent to the first request.
GET https://example.org/forum/questions/
[Query]
search: Install Linux
order: newest
```

> Query parameters in query parameter section are not URL encoded.

When query parameters are present in the URL and in a query parameters section, the resulting request will have both parameters.

Headers

Optional list of HTTP request headers.

A header consists of a name, followed by a `:` and a value.

```
GET https://example.org/news
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.14; rv:70.0)
Gecko/20100101 Firefox/70.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
```

> Headers directly follow URL, without any section name, contrary to query parameters, form parameters or cookies

Note that a header usually doesn't start with double quotes. If a header value starts with double quotes, double quotes will be part of the header value:

```
PATCH https://example.org/file.txt
If-Match: "e0023aa4e"
```

`If-Match` request header will be sent will the following value `"e0023aa4e"` (started and ended with double quotes).

Headers must follow directly after the method and URL.

Options

Options used to execute this request.

Options such as `--location`, `--verbose`, `--insecure` can be used at the command line and applied to every request of an Hurl file. An `[Options]` section can be used to apply option to only one request (without passing options to the command line), while other requests are unaffected.

```
GET https://example.org
# An options section, each option is optional and applied only to this
```

```
request...
[Options]
aws-sigv4: aws:amz:sts     # generate AWS SigV4 Authorization header
cacert: /etc/cert.pem      # custom certificate file
cert: /etc/client-cert.pem # client authentication certificate
key: /etc/client-cert.key  # client authentication certificate key
compressed: true           # request a compressed response
connect-timeout: 20s       # connect timeout
delay: 3s                  # delay for this request (aka sleep)
fail-with-body: true       # Output HTTP response even if there are assert
errors
http3: true                # use HTTP/3 protocol version
insecure: true             # allow insecure SSL connections and transfers
ipv6: true                 # use IPv6 addresses
limit-rate: 32000          # limit this request to the specidied speed
(bytes/s)
location: true             # follow redirection for this request
max-redirs: 10             # maximum number of redirections
max-time: 30s              # maximum time for a request/response
no-header: Accept          # header name to remove from the request
no-jsonpath-coercion: true # disable JSONPath result coercion for this request
output: out.html           # dump the response to this file
path-as-is: true           # do not handle sequences of /../ or /./ in URL path
retry: 10                  # number of retry if HTTP/asserts errors
retry-interval: 500ms      # interval between retry
skip: false                # skip this request
tls-max: 1.2               # maximum TLS version
unix-socket: sock          # use Unix socket for transfer
user: bob:secret           # use basic authentication
proxy: my.proxy:8012       # define proxy (host:port where host can be an IP
address)
variable: country=Italy    # define variable country
variable: planet=Earth     # define variable planet
verbose: true              # allow verbose output
very-verbose: true         # allow more verbose output
```

> Variables defined in an `[Options]` section, including values loaded from `variables-file`, are defined also for the next entries. This is the exception, all other options are defined only for the current request.

Query parameters

Optional list of query parameters.

A query parameter consists of a field, followed by a `:` and a value. The query parameters section starts with `[Query]`. Contrary to query parameters in the URL, each value in the query parameters section is not URL encoded.

```
GET https://example.org/news
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.14; rv:70.0)
Gecko/20100101 Firefox/70.0
[Query]
```

```
order: newest
search: {{custom-search}}
count: 100
```

If there are any parameters in the URL, the resulted request will have both parameters.

Form parameters

A form parameters section can be used to send data, like HTML form.

This section contains an optional list of key values, each key followed by a `:` and a value. Key values will be encoded in key-value tuple separated by '&', with a '=' between the key and the value, and sent in the body request. The content type of the request is `application/x-www-form-urlencoded`. The form parameters section starts with `[Form]`.

```
POST https://example.org/contact
[Form]
default: false
token: {{token}}
email: john.doe@rookie.org
number: 33611223344
```

Form parameters section can be seen as syntactic sugar over body section (values in form parameters section are not URL encoded.). A oneline string body could be used instead of a forms parameters section.

```
# Run a POST request with form parameters section:
POST https://example.org/test
[Form]
name: John Doe
key1: value1

# Run the same POST request with a body section:
POST https://example.org/test
Content-Type: application/x-www-form-urlencoded
`name=John%20Doe&key1=value1`
```

When both body section and form parameters section are present, only the body section is taken into account.

Multipart Form Data

A multipart form data section can be used to send data, with key / value and file content (see multipart/form-data on MDN).

The form parameters section starts with `[Multipart]`.

```
POST https://example.org/upload
[Multipart]
field1: value1
```

```
field2: file,example.txt;
# One can specify the file content type:
field3: file,example.zip; application/zip
```

Files are relative to the input Hurl file, and cannot contain implicit parent directory (`..`). You can use `--file-root` option to specify the root directory of all file nodes.

Content type can be specified or inferred based on the filename extension:

- `.gif`: `image/gif`,
- `.jpg`: `image/jpeg`,
- `.jpeg`: `image/jpeg`,
- `.png`: `image/png`,
- `.svg`: `image/svg+xml`,
- `.txt`: `text/plain`,
- `.htm`: `text/html`,
- `.html`: `text/html`,
- `.pdf`: `application/pdf`,
- `.xml`: `application/xml`

By default, content type is `application/octet-stream`.

As an alternative to a `[Multipart]` section, multipart forms can also be sent with a multiline string body:

```
POST https://example.org/upload
Content-Type: multipart/form-data; boundary="boundary"
```
--boundary
Content-Disposition: form-data; name="key1"

value1
--boundary
Content-Disposition: form-data; name="upload1"; filename="data.txt"
Content-Type: text/plain

Hello World!
--boundary
Content-Disposition: form-data; name="upload2"; filename="data.html"
Content-Type: text/html

Hello World!
--boundary--
```
```

> When using a multiline string body to send a multipart form data, files content must be inlined in the Hurl file.

Cookies

Optional list of session cookies for this request.

A cookie consists of a name, followed by a `:` and a value. Cookies are sent per request, and are not added to the cookie storage session, contrary to a cookie set in a header response. (for instance `Set-Cookie: theme=light`). The cookies section starts with `[Cookies]`.

```
GET https://example.org/index.html
[Cookies]
theme: light
sessionToken: abc123
```

Cookies section can be seen as syntactic sugar over corresponding request header.

```
# Run a GET request with cookies section:
GET https://example.org/index.html
[Cookies]
theme: light
sessionToken: abc123

# Run the same GET request with a header:
GET https://example.org/index.html
Cookie: theme=light; sessionToken=abc123
```

Basic Authentication

A basic authentication section can be used to perform basic authentication.

Username is followed by a `:` and a password. The basic authentication section starts with `[BasicAuth]`. Username and password are *not* base64 encoded.

```
# Perform basic authentication with login `bob` and password `secret`.
GET https://example.org/protected
[BasicAuth]
bob: secret
```

> Spaces surrounded username and password are trimmed. If you really want a space in your password (!!), you could use Hurl unicode literals .

This is equivalent (but simpler) to construct the request with a Authorization header:

```
# Authorization header value can be computed with `echo -n 'bob:secret' |
base64`
GET https://example.org/protected
Authorization: Basic Ym9iOnNlY3JldA==
```

Basic authentication allows per request authentication. If you want to add basic authentication to all the requests of a Hurl file you can use `-u/--user` option.

Body

Optional HTTP body request.

If the body of the request is a JSON string or a XML string, the value can be directly inserted without any modification. For a text based body that is neither JSON nor XML, one can use multiline string body that starts with ``` and ends with ```. Multiline string body support "language hint" and can be used to create GraphQL queries.

For a precise byte control of the request body, Base64 encoded string, hexadecimal string or included file can be used to describe exactly the body byte content.

> You can set a body request even with a `GET` body, even if this is not a common practice.

The body section must be the last section of the request configuration.

JSON body

JSON request body is used to set a literal JSON as the request body.

```
# Create a new doggy thing with JSON body:
POST https://example.org/api/dogs
{
    "id": 0,
    "name": "Frieda",
    "picture": "images/scottish-terrier.jpeg",
    "age": 3,
    "breed": "Scottish Terrier",
    "location": "Lisco, Alabama"
}
```

JSON request body can be templatized with variables:

```
# Create a new catty thing with JSON body:
POST https://example.org/api/cats
{
    "id": 42,
    "lives": {{lives_count}},
    "name": "{{ name }}"
}
```

When using JSON request body, the content type `application/json` is automatically set.

JSON request body can be seen as syntactic sugar of multiline string body with `json` identifier:

```
# Create a new doggy thing with JSON body:
POST https://example.org/api/dogs
```json
{
    "id": 0,
    "name": "Frieda",
    "picture": "images/scottish-terrier.jpeg",
    "age": 3,
```
```

```
    "breed": "Scottish Terrier",
    "location": "Lisco, Alabama"
}
```

If you don't want templates to be evaluated inside JSON body you can use multiline string body with `raw` identifier:

```
# {{name}} is not a variable
POST https://example.org/api/cats
Content-Type: application/json
```raw
{
  "id": 42,
  "name": "{{ name }}"
}
```
```

XML body

XML request body is used to set a literal XML as the request body.

```
# Create a new soapy thing XML body:
POST https://example.org/InStock
Content-Type: application/soap+xml; charset=utf-8
Content-Length: 299
SOAPAction: "http://www.w3.org/2003/05/soap-envelope"
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:m="http://example.net">
  <soap:Header></soap:Header>
  <soap:Body>
    <m:GetStockPrice>
      <m:StockName>GOOG</m:StockName>
    </m:GetStockPrice>
  </soap:Body>
</soap:Envelope>
```

XML request body can be seen as syntactic sugar of multiline string body with `xml` identifier:

```
# Create a new soapy thing XML body:
POST https://example.org/InStock
Content-Type: application/soap+xml; charset=utf-8
Content-Length: 299
SOAPAction: "http://www.w3.org/2003/05/soap-envelope"
```xml
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:m="http://example.net">
  <soap:Header></soap:Header>
```
```

```
  <soap:Body>
    <m:GetStockPrice>
      <m:StockName>GOOG</m:StockName>
    </m:GetStockPrice>
  </soap:Body>
</soap:Envelope>
```

> Contrary to JSON body, the succinct syntax of XML body can not use variables. If you need to use variables in your XML body, use a simple multiline string body with variables.

GraphQL query

GraphQL query uses multiline string body with `graphql` identifier:

```
POST https://example.org/starwars/graphql
```graphql
{
  human(id: "1000") {
    name
    height(unit: FOOT)
  }
}
```
```

GraphQL query body can use GraphQL variables:

```
POST https://example.org/starwars/graphql
```graphql
query Hero($episode: Episode, $withFriends: Boolean!) {
  hero(episode: $episode) {
    name
    friends @include(if: $withFriends) {
      name
    }
  }
}

variables {
  "episode": "JEDI",
  "withFriends": false
}
```
```

GraphQL query, as every multiline string body, can use Hurl variables.

```
POST https://example.org/starwars/graphql
```graphql
{
  human(id: "{{human_id}}") {
```
```

```
    name
    height(unit: FOOT)
  }
}
```

> Hurl variables and GraphQL variables can be mixed in the same body.

Multiline string body

For text based body that are neither JSON nor XML, one can use multiline string, started and ending with ```.

```
POST https://example.org/models
```
Year,Make,Model,Description,Price
1997,Ford,E350,"ac, abs, moon",3000.00
1999,Chevy,"Venture ""Extended Edition""","",4900.00
1999,Chevy,"Venture ""Extended Edition, Very Large""",,5000.00
1996,Jeep,Grand Cherokee,"MUST SELL! air, moon roof, loaded",4799.00
```
```

The standard usage of a multiline string is:

```
```
line1
line2
line3
```
```

is evaluated as “line1”.

Multiline string body can be templatized with variables:

```
POST https://example.org/models
[Options]
variable: var1=lemon
variable: var2=yellow
```
Fruit,Color
{{var1}},{{var2}}
```
```

Escapes are not processed (i.e. Hurl Unicode literals are not supported): `\n` is two consecutive chars (\ followed by a `n`), not a single newline char.

Multiline string body can use language identifier, like `json`, `xml`, `graphql` or `raw`. Depending on the language identifier, an additional ‘Content-Type’ request header is sent, and the real body (bytes sent over the wire) can be different from the raw multiline text.

````
POST https://example.org/api/dogs
```json
{
    "id": 0,
    "name": "Frieda"
}
```
````

Raw multiline string body don't evaluate templates:

````
# {{name}} is not a variable
POST https://example.org/api/cats
Content-Type: application/json
```raw
{
    "id": 42,
    "lives": {{ lives_count }},
    "name": "{{ name }}"
}
```
````

Oneline string body

For text based body that do not contain newlines, one can use oneline string, started and ending with `.

```
POST https://example.org/helloworld
`Hello world!`
```

Base64 body

Base64 body is used to set binary data as the request body.

Base64 body starts with `base64,` and end with `;`. MIME's Base64 encoding is supported (newlines and white spaces may be present anywhere but are to be ignored on decoding), and = padding characters might be added.

```
POST https://example.org
# Some random comments before body
base64,TG9yZW0gaXBzdW0gZG9sb3Igc2l0IGFtZXQsIGNvbnNlY3RldHVyIG
FkaXBpc2NpbmcgZWxpdC4gSW4gbWFsZXN1YWRhLCBuaXNsIHZlbCBkaWN0dW0g
aGVuZHJlcml0LCBlc3QganVzdG8gYmliZW5kdW0gbWV0dXMsIG5lYyBydXRydW
0gdG9ydG9yIG1hc3NhIGlkIG1ldHVzLiA=;
```

Hex body

Hex body is used to set binary data as the request body.

Hex body starts with `hex,` and end with `;`.

```
PUT https://example.org
# Send a café, encoded in UTF-8
hex,636166c3a90a;
```

File body

To use the binary content of a local file as the body request, file body can be used. File body starts with `file,` and ends with `;``

```
POST https://example.org
# Some random comments before body
file,data.bin;
```

File are relative to the input Hurl file, and cannot contain implicit parent directory (`..`). You can use `--file-root` option to specify the root directory of all file nodes.

Response

Definition

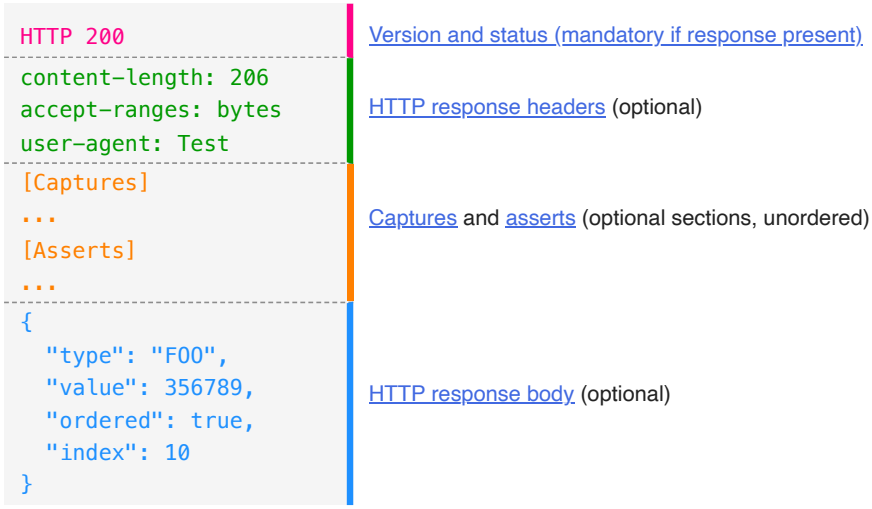
Responses can be used to capture values to perform subsequent requests, or add asserts to HTTP responses. Response on requests are optional, a Hurl file can just consist of a sequence of requests.

A response describes the expected HTTP response, with mandatory version and status, followed by optional headers, captures, asserts and body. Assertions in the expected HTTP response describe values of the received HTTP response. Captures capture values from the received HTTP response and populate a set of named variables that can be used in the following entries.

Example

```
GET https://example.org
HTTP 200
Last-Modified: Wed, 21 Oct 2015 07:28:00 GMT
[Asserts]
xpath "normalize-space(//head/title)" startsWith "Welcome"
xpath "//li" count == 18
```

Structure



Capture and Assertion

With the response section, one can optionally capture value from headers, body, or add assert on status code, body or headers.

Body compression

Hurl outputs the raw HTTP body to stdout by default. If response body is compressed (using br, gzip, deflate), the binary stream is output, without any modification. One can use `--compressed` option to request a compressed response and automatically get the decompressed body.

Captures and asserts work automatically on the decompressed body, so you can request compressed data (using `Accept-Encoding` header by example) and add assert and captures on the decoded body as if there weren't any compression.

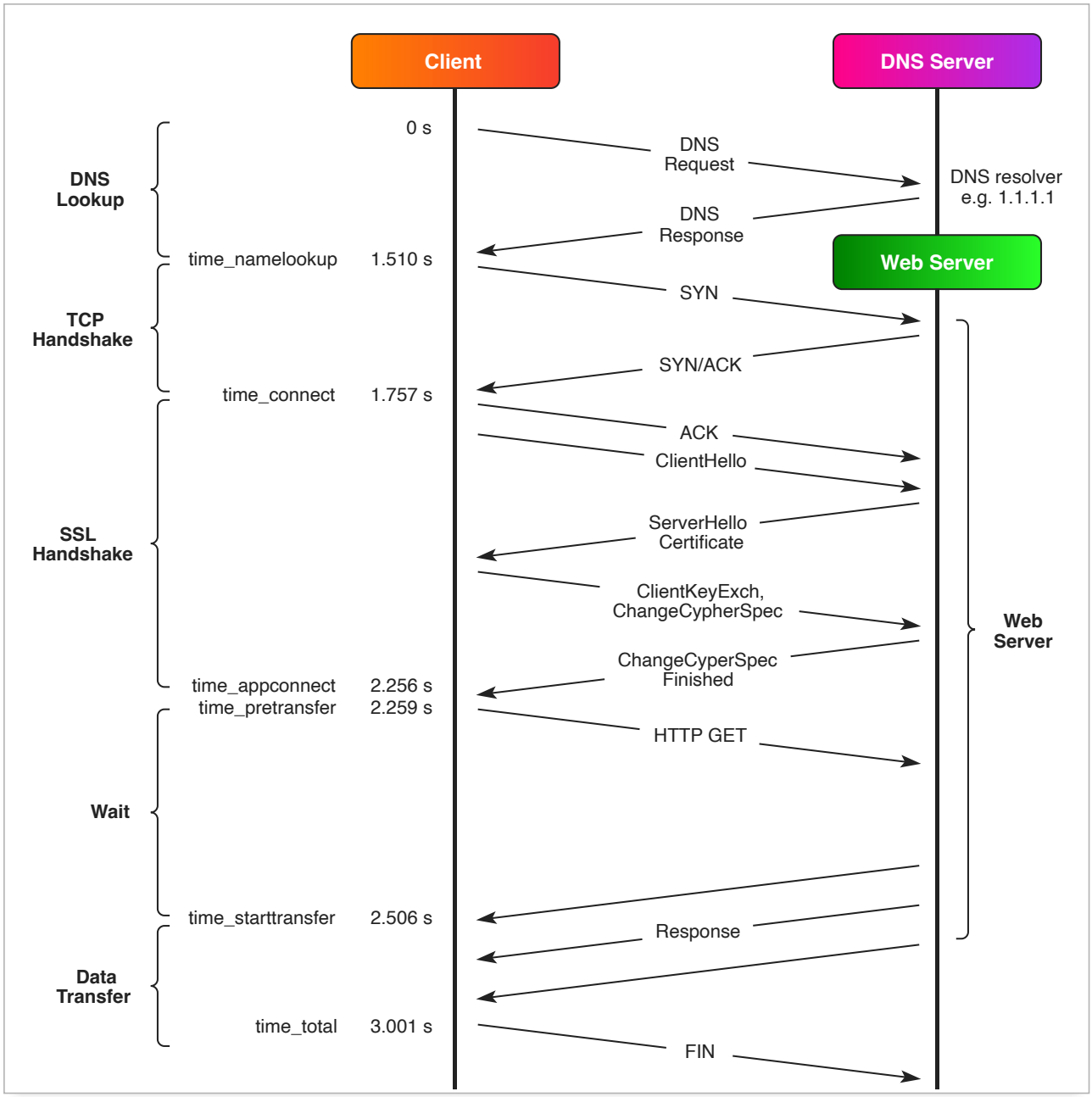
Timings

HTTP response timings are exposed through Hurl structured output (see `--json`), HTML report (see `--report-html`) and JSON report (see `--report-json`).

On each response, libcurl response timings are available:

- **time_namelookup**: the time it took from the start until the name resolving was completed. You can use `--resolve` to exclude DNS performance from the measure.
- **time_connect**: The time it took from the start until the TCP connect to the remote host (or proxy) was completed.
- **time_appconnect**: The time it took from the start until the SSL/SSH/etc connect/handshake to the remote host was completed. The client is then ready to send its HTTP GET request.
- **time_starttransfer**: The time it took from the start until the first byte was just about to be transferred (just before Hurl reads the first byte from the network). This includes time_pretransfer and also the time the server needed to calculate the result.
- **time_total**: The total time that the full operation lasted.

All timings are in microsecond.



Courtesy of Cloudflare

Capturing Response

Captures

Captures are optional values that are **extracted from the HTTP response** and stored in a named variable. These captures may be the response status code, part of or the entire the body, and response headers.

Captured variables can be accessed through a run session; each new value of a given variable overrides the last value.

Captures can be useful for using data from one request in another request, such as when working with CSRF tokens. Variables in a Hurl file can be created from captures or injected into the session.

```
# An example to show how to pass a CSRF token
# from one request to another:

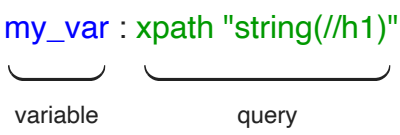
# First GET request to get CSRF token value:
GET https://example.org
HTTP 200
# Capture the CSRF token value from html body.
[Captures]
csrf_token: xpath "normalize-space(//meta[@name='_csrf_token']/@content)"

# Do the login !
POST https://acmecorp.net/login?user=toto&password=1234
X-CSRF-TOKEN: {{csrf_token}}
HTTP 302
```

Body responses can be encoded by server (see [`Content-Encoding` HTTP header][content-encoding](https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Content-Encoding)) but captures in Hurl files are not affected by this content compression. All body captures (`body`, `bytes`, `sha256` etc…) except `rawbytes` work *after* content decoding.

Finally, body text captures (`body`, `jsonpath`, `xpath` etc…) are also decoded to strings based on `Content-Type` header so these queries can be captures as usual strings.

Structure of a capture:



A capture consists of a variable name, followed by `:` and a query. Captures section starts with `[Captures]`.

Query

Queries are used to extract data from an HTTP response.

A query can extract data from

- status line:
  - `status`
  - `version`
- headers:
  - `header`
  - `cookie`
- body:
  - `body`
  - `bytes`
  - `rawbytes`
  - `xpath`
  - `jsonpath`
  - `regex`
  - `sha256`
  - `md5`
- others:
  - `url`
  - `redirects`
  - `ip`
  - `variable`
  - `duration`
  - `certificate`

Extracted data can then be further refined using filters.

Status capture

Capture the received HTTP response status code. Status capture consists of a variable name, followed by a `:`, and the keyword `status`.

```
GET https://example.org
HTTP 200
[Captures]
my_status: status
```

Version capture

Capture the received HTTP version. Version capture consists of a variable name, followed by a `:`, and the keyword `version`. The value captured is a string:

```
GET https://example.org
HTTP 200
[Captures]
http_version: version
```

Header capture

Capture a header from the received HTTP response headers. Header capture consists of a variable name, followed by a `:`, then the keyword `header` and a header name.

```
POST https://example.org/login
[Form]
user: toto
password: 12345678
HTTP 302
[Captures]
next_url: header "Location"
```

Cookie capture

Capture a `Set-Cookie` header from the received HTTP response headers. Cookie capture consists of a variable name, followed by a `:`, then the keyword `cookie` and a cookie name.

```
GET https://example.org/cookies/set
HTTP 200
[Captures]
session-id: cookie "LSID"
```

Cookie attributes value can also be captured by using the following format: `<cookie-name>[cookie-attribute]`. The following attributes are supported: `Value`, `Expires`, `Max-Age`, `Domain`, `Path`, `Secure`, `HttpOnly` and `SameSite`.

```
GET https://example.org/cookies/set
HTTP 200
[Captures]
value1: cookie "LSID"
value2: cookie "LSID[Value]"     # Equivalent to the previous capture
expires: cookie "LSID[Expires]"
max-age: cookie "LSID[Max-Age]"
domain: cookie "LSID[Domain]"
path: cookie "LSID[Path]"
secure: cookie "LSID[Secure]"
http-only: cookie "LSID[HttpOnly]"
same-site: cookie "LSID[SameSite]"
```

Body capture

Capture the entire body (decoded as text) from the received HTTP response. The encoding used to decode the body is based on the `charset` value in the `Content-Type` header response.

```
GET https://example.org/home
HTTP 200
[Captures]
my_body: body
```

If the `Content-Type` doesn't include any encoding hint, a decode filter can be used to explicitly decode the body response bytes.

```
# Our HTML response is encoded using GB 2312.
# But, the 'Content-Type' HTTP response header doesn't precise any charset,
# so we decode explicitly the bytes.
GET https://example.org/cn
HTTP 200
[Captures]
my_body: bytes charsetDecode "gb2312"
```

`body` capture works *after* content encoding decompression (so the captured value is not affected by `Content-Encoding` response header).

Bytes capture

Capture the entire body (as a raw bytestream) from the received HTTP response

```
GET https://example.org/data.bin
HTTP 200
[Captures]
my_data: bytes
```

Like `body` capture, `bytes` capture works *after* content encoding decompression (so the captured value is not affected by `Content-Encoding` response header).

RawBytes capture

Capture the entire body as a raw bytestream from the received HTTP response, *before* any content decoding.

```
GET https://example.org/data.bin
HTTP 200
[Captures]
raw_data: rawbytes
```

Unlike `bytes` capture, `rawbytes` returns the raw bytes before content encoding decompression. For uncompressed responses, `rawbytes` and `bytes` capture the same data.

XPath capture

Capture a XPath query from the received HTTP body decoded as a string. Currently, only XPath 1.0 expression can be used.

```
GET https://example.org/home
# Capture the identifier from the dom node <div id="pet0">5646eaf23</div
HTTP 200
[Captures]
pet-id: xpath "normalize-space(//div[@id='pet0'])"

# Open the captured page.
GET https://example.org/home/pets/{{pet-id}}
HTTP 200
```

XPath captures are not limited to node values (like string, or boolean); any valid XPath can be captured and asserted with variable asserts.

```
# Test that the XML endpoint return 200 pets
GET https://example.org/api/pets
HTTP 200
[Captures]
pets: xpath "//pets"
[Asserts]
variable "pets" count == 200
```

XPath expression can also be evaluated against part of the body with a `xpath` filter:

```
GET https://example.org/home_cn
HTTP 200
[Captures]
pet-id: bytes charsetDecode "gb2312" xpath "normalize-space(//div[@id='pet0'])"
```

JSONPath capture

Capture a JSONPath query from the received HTTP body.

```
POST https://example.org/api/contact
[Form]
token: {{token}}
email: toto@rookie.net
HTTP 200
[Captures]
contact-id: jsonpath "$['id']"
```

> Explain that the value selected by the JSONPath is coerced to a string when only one node is selected.

As with XPath captures, JSONPath captures can be anything from string, number, to object and collections. For instance, if we have a JSON endpoint that returns the following JSON:

```
{
  "a_null": null,
  "an_object": {
    "id": "123"
  },
  "a_list": [
    1,
    2,
    3
  ],
  "an_integer": 1,
  "a float": 1.1,
  "a_bool": true,
  "a_string": "hello"
}
```

We can capture the following paths:

```
GET https://example.org/captures-json
HTTP 200
[Captures]
an_object:  jsonpath "$['an_object']"
a_list:     jsonpath "$['a_list']"
a_null:     jsonpath "$['a_null']"
an_integer: jsonpath "$['an_integer']"
a_float:    jsonpath "$['a_float']"
a_bool:     jsonpath "$['a_bool']"
a_string:   jsonpath "$['a_string']"
all:        jsonpath "$"
```

Regex capture

Capture a regex pattern from the HTTP received body, decoded as text.

```
GET https://example.org/helloworld
HTTP 200
[Captures]
id_a: regex "id_a:([0-9]+)"
id_b: regex "id_b:(\\d+)"   # pattern using double quote
id_c: regex /id_c:(\d+)/    # pattern using forward slash
name: regex "Hello ([a-zA-Z]+)"
```

The regex pattern must have at least one capture group, otherwise the capture will fail. When the pattern is a double-quoted string, metacharacters beginning with a backslash in the pattern (like `\d`, `\s`) must be escaped; literal pattern enclosed by `/` can also be used to avoid metacharacters escaping.

The regex syntax is documented at https://docs.rs/regex/latest/regex/#syntax. For instance, one can use flags to enable case-insensitive match:

```
GET https://example.org/hello
HTTP 200
[Captures]
word: regex /(?i)hello (\w+)!/
```

SHA-256 capture

Capture the SHA-256 hash of the response body.

```
GET https://example.org/data.tar.gz
HTTP 200
[Captures]
my_hash: sha256
```

Like `body` assert, `sha256` capture works *after* content encoding decompression (so the captured value is not affected by `Content-Encoding` response header).

MD5 capture

Capture the MD5 hash of the response body.

```
GET https://example.org/data.tar.gz
HTTP 200
[Captures]
my_hash: md5
```

Like `sha256` asserts, `md5` assert works *after* content encoding decompression (so the predicates values are not affected by `Content-Encoding` response header)

URL capture

Capture the last fetched URL. This is most meaningful if you have told Hurl to follow redirection (see `[Options]` section(#file-format-request-options) or `--location` option). URL capture consists of a variable name, followed by a `:`, and the keyword `url`.

```
GET https://example.org/redirecting
[Options]
location: true
HTTP 200
[Captures]
landing_url: url
```

Redirects capture

Capture each step of redirection. This is most meaningful if you have told Hurl to follow redirection (see `[Options]`section(#file-format-request-options) or `--location` option). Redirects capture

consists of a variable name, followed by a `:`, and the keyword `redirects`. Redirects query returns a collection so each step of the redirection can be capture.

```
GET https://example.org/redirecting/1
[Options]
location: true
HTTP 200
[Asserts]
redirects count == 3
[Captures]
step1: redirects nth 0 location
step2: redirects nth 1 location
step3: redirects nth 2 location
```

IP address capture

Capture the IP address of the last connection. The value of the `ip` query is a string.

```
GET https://example.org/hello
HTTP 200
[Captures]
server_ip: ip
```

Variable capture

Capture the value of a variable into another.

```
GET https://example.org/helloworld
HTTP 200
[Captures]
in: body
name: variable "in"
```

Duration capture

Capture the response time of the request in ms.

```
GET https://example.org/helloworld
HTTP 200
[Captures]
duration_in_ms: duration
```

SSL certificate capture

Capture the SSL certificate properties. Certificate capture consists of the keyword `certificate`, followed by the certificate attribute value.

The following attributes are supported: `Subject`, `Issuer`, `Start-Date`, `Expire-Date`, `Serial-Number`, `Subject-Alt-Name` and `Value`.

```
GET https://example.org
HTTP 200
[Captures]
cert_subject: certificate "Subject"
cert_issuer: certificate "Issuer"
cert_expire_date: certificate "Expire-Date"
cert_serial_number: certificate "Serial-Number"
san: certificate "Serial-Number"
value: certificate "Value"
```

Redacting Secrets

When capturing data, you may need to hide captured values from logs and report. To do this, captures can use secrets which are redacted from logs and reports, using `--secret` option:

```
$ hurl --secret pass=sesame-ouvre-toi file.hurl
```

If the secret value to be redacted is dynamic, or not known before execution, a capture can become a secret using `redact` at the end of the query's capture:

```
GET https://foo.com
HTTP 200
[Captures]
pass: header "token" redact
```

Asserting Response

Asserts

Asserts are used to test various properties of an HTTP response. Asserts can be implicits (such as version, status, headers) or explicit within an `[Asserts]` section. The delimiter of the request / response is `HTTP <STATUS-CODE>`: after this delimiter, you'll find the implicit asserts, then an `[Asserts]` section with all the explicit checks.

```
GET https://example.org/api/cats
HTTP 200
# Implicit assert on `Content-Type` Header
Content-Type: application/json; charset=utf-8
[Asserts]
# Explicit asserts section
bytes count == 120
header "Content-Type" contains "utf-8"
jsonpath "$.cats" count == 49
jsonpath "$.cats[0].name" == "Felix"
jsonpath "$.cats[0].lives" == 9
```

Body responses can be encoded by server (see [`Content-Encoding` HTTP header][content-encoding](https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Content-Encoding)) but asserts in Hurl files are not affected by this content compression. All body asserts (`body`, `bytes`, `sha256` etc…) except `rawbytes` work *after* content decoding.

Finally, body text asserts (`body`, `jsonpath`, `xpath` etc…) are also decoded to strings based on `Content-Type` header so these asserts can be written with usual strings.

Structure

The asserts order in a Hurl file is:

- implicit asserts on version and status
- implicit asserts on headers
- explicit asserts
- implicit assert on body

```
HTTP 200
```
Version and status (mandatory if response present)

```
content-length: 206
accept-ranges: bytes
user-agent: Test
```
HTTP response headers (optional)

```
[Captures]
...
[Asserts]
...
```
Captures and explicit asserts (optional sections, unordered)

```
{
  "type": "FOO",
  "value": 356789,
  "ordered": true,
  "index": 10
}
```

HTTP response body (optional)

Implicit asserts

Version - Status

Expected protocol version and status code of the HTTP response.

Protocol version is one of `HTTP/1.0`, `HTTP/1.1`, `HTTP/2`, `HTTP/3` or `HTTP`; `HTTP` describes any version. Note that there are no status text following the status code.

```
GET https://example.org/404.html
HTTP 404
```

Wildcard keywords `HTTP` and `*` can be used to disable tests on protocol version and status:

```
GET https://example.org/api/pets
HTTP *
# Check that response status code is > 400 and <= 500
[Asserts]
status > 400
status <= 500
```

While `HTTP/1.0`, `HTTP/1.1`, `HTTP/2` and `HTTP/3` explicitly check HTTP version:

```
# Check that our server responds with HTTP/2
GET https://example.org/api/pets
HTTP/2 200
```

Headers

Optional list of the expected HTTP response headers that must be in the received response.

A header consists of a name, followed by a `:` and a value.

For each expected header, the received response headers are checked. If the received header is not equal to the expected, or not present, an error is raised. The comparison is case-insensitive for the name: expecting a `Content-Type` header is equivalent to a `content-type` one. Note that the expected headers list is not fully descriptive: headers present in the response and not in the expected list doesn't raise error.

```
# Check that user toto is redirected to home after login.
POST https://example.org/login
```

```
[Form]
user: toto
password: 12345678
HTTP 302
Location: https://example.org/home
```

> Quotes in the header value are part of the value itself.
>
> This is used by the ETag Header
>
> ```
> ETag: W/"<etag_value>"
> ETag: "<etag_value>"
> ```

Testing duplicated headers is also possible.

For example with the `Set-Cookie` header:

```
Set-Cookie: theme=light
Set-Cookie: sessionToken=abc123; Expires=Wed, 09 Jun 2021 10:18:14 GMT
```

You can either test the two header values:

```
GET https://example.org/index.html
Host: example.net
HTTP 200
Set-Cookie: theme=light
Set-Cookie: sessionToken=abc123; Expires=Wed, 09 Jun 2021 10:18:14 GMT
```

Or only one:

```
GET https://example.org/index.html
Host: example.net
HTTP 200
Set-Cookie: theme=light
```

If you want to test specifically the number of headers returned for a given header name, or if you want to test header value with predicates (like `startsWith`, `contains`, `exists`)you can use the explicit header assert.

Body

Optional assertion on the received HTTP response body. Body section can be seen as syntactic sugar over body asserts (with == predicate). If the body of the response is a JSON string or a XML string, the body assertion can be directly inserted without any modification. For a text based body that is neither JSON nor XML, one can use multiline string that starts with ``` and ends with ```. For a precise byte control of the response body, a Base64 encoded string or an input file can be used to describe exactly the body byte content to check.

Like explicit body assert, the body section is automatically decompressed based on the value of `Content-Encoding` response header. So, whatever is the response compression (`gzip`, `brotli`, etc…) body section doesn't depend on the content encoding. For textual body sections (JSON, XML, multiline, etc…), content is also decoded to string, based on the value of `Content-Type` response header.

JSON body

```
# Get a doggy thing:
GET https://example.org/api/dogs/{{dog-id}}
HTTP 200
{
    "id": 0,
    "name": "Frieda",
    "picture": "images/scottish-terrier.jpeg",
    "age": 3,
    "breed": "Scottish Terrier",
    "location": "Lisco, Alabama"
}
```

JSON response body can be seen as syntactic sugar of multiline string body with `json` identifier:

```
# Get a doggy thing:
GET https://example.org/api/dogs/{{dog-id}}
HTTP 200
```json
{
    "id": 0,
    "name": "Frieda",
    "picture": "images/scottish-terrier.jpeg",
    "age": 3,
    "breed": "Scottish Terrier",
    "location": "Lisco, Alabama"
}
```
```

XML body

```
GET https://example.org/api/catalog
HTTP 200
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
   <book id="bk101">
      <author>Gambardella, Matthew</author>
      <title>XML Developer's Guide</title>
      <genre>Computer</genre>
      <price>44.95</price>
      <publish_date>2000-10-01</publish_date>
      <description>An in-depth look at creating applications with XML.
</description>
```

````
   </book>
</catalog>
````

XML response body can be seen as syntactic sugar of multiline string body with `xml` identifier:

````
GET https://example.org/api/catalog
HTTP 200
```xml
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
   <book id="bk101">
      <author>Gambardella, Matthew</author>
      <title>XML Developer's Guide</title>
      <genre>Computer</genre>
      <price>44.95</price>
      <publish_date>2000-10-01</publish_date>
      <description>An in-depth look at creating applications with XML.
</description>
   </book>
</catalog>
```
````

Multiline string body

````
GET https://example.org/models
HTTP 200
```
Year,Make,Model,Description,Price
1997,Ford,E350,"ac, abs, moon",3000.00
1999,Chevy,"Venture ""Extended Edition""","",4900.00
1999,Chevy,"Venture ""Extended Edition, Very Large""",,5000.00
1996,Jeep,Grand Cherokee,"MUST SELL! air, moon roof, loaded",4799.00
```
````

The standard usage of a multiline string is :

````
```
line1
line2
line3
```
````

Oneline string body

For text based response body that do not contain newlines, one can use oneline string, started and ending with `.

````
POST https://example.org/helloworld
HTTP 200
````

```
`Hello world!`
```

Base64 body

Base64 response body assert starts with `base64,` and end with `;`. MIME's Base64 encoding is supported (newlines and white spaces may be present anywhere but are to be ignored on decoding), and = padding characters might be added.

```
GET https://example.org
HTTP 200
base64,TG9yZW0gaXBzdW0gZG9sb3Igc2l0IGFtZXQsIGNvbnNlY3RldHVyIG
FkaXBpc2NpbmcgZWxpdC4gSW4gbWFsZXN1YWRhLCBuaXNsIHZlbCBkaWN0dW0g
aGVuZHJlcml0LCBlc3QganVzdG8gYmliZW5kdW0gbWV0dXMsIG5lYyBydXRydW
0gdG9ydG9yIG1hc3NhIGlkIG1ldHVzLiA=;
```

File body

To use the binary content of a local file as the body response assert, file body can be used. File body starts with `file,` and ends with `;``

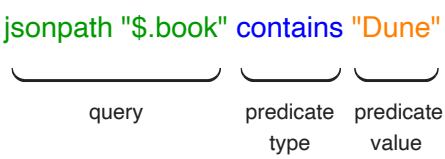
```
GET https://example.org
HTTP 200
file,data.bin;
```

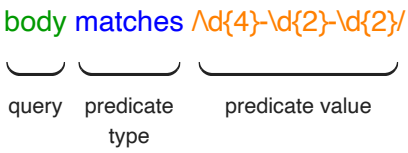
File are relative to the input Hurl file, and cannot contain implicit parent directory (`..`). You can use `--file-root` option to specify the root directory of all file nodes.

Explicit asserts

Optional list of assertions on the HTTP response within an `[Asserts]` section. Assertions can describe checks on status code, on the received body (or part of it) and on response headers.

Structure of an explicit assert:





An assert consists of a query followed by a predicate. The format of the query is shared with captures, and queries can extract data from

- status line:
  - status
  - version
- headers:
  - header
  - cookie
- body:
  - body
  - bytes
  - rawbytes
  - xpath
  - jsonpath
  - regex
  - sha256
  - md5
- others:
  - url
  - redirects
  - ip
  - variable
  - duration
  - certificate

Queries, in asserts and in captures, can be refined with filters, like count(#file-format-filters-count) to add tests on collections sizes.

Predicates

Predicates consist of a predicate function and a predicate value. Predicate functions are:

Predicate	Description	Example
==	Query and predicate value are equal	`jsonpath "$.book" == "Dune"`
!=	Query and predicate value are different	`jsonpath "$.color" != "red"`

>	Query number or date is greater than predicate value	`jsonpath "$.year" > 1978`
		
		`jsonpath "$.createdAt" toDate "%+" > {{ a_date }}`

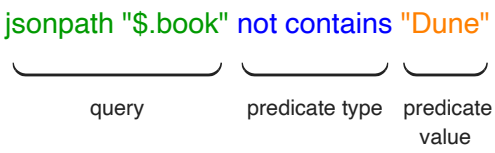
Predicate	Description	Example
`>=`	Query number or date is greater than or equal to the predicate value	`jsonpath "$.year" >= 1978`
`<`	Query number or date is less than that predicate value	`jsonpath "$.year" < 1978`
`<=`	Query number or date is less than or equal to the predicate value	`jsonpath "$.year" <= 1978`
`startsWith`	Query starts with the predicate value Value is string or a binary content	`jsonpath "$.movie" startsWit` `bytes startsWith hex,efbbbf;`
`endsWith`	Query ends with the predicate value Value is string or a binary content	`jsonpath "$.movie" endsWith` `bytes endsWith hex,ab23456;`

`contains`	If query returns a collection of string or numbers, query collection includes the predicate value (string or number)
	If query returns a string or a binary content, query contains the predicate value (string or bytes)
		`jsonpath "$.movie" contains`
		`bytes contains hex,beef;`
		`jsonpath "$.numbers" contain`

`matches`	Part of the query string matches the regex pattern described by the predicate value (see regex syntax)	`jsonpath "$.release" matches` `jsonpath "$.release" matches`
`exists`	Query returns a value	`jsonpath "$.book" exists`
`isBoolean`	Query returns a boolean	`jsonpath "$.succeeded" isBoo`
`isEmpty`	Query returns an empty collection (list, object)	`jsonpath "$.movies" isEmpty`
`isFloat`	Query returns a float	`jsonpath "$.height" isFloat`
`isInteger`	Query returns an integer	`jsonpath "$.count" isInteger`
`isIpv4`	Query returns an IPv4 address	`ip isIpv4`
`isIpv6`	Query returns an IPv6 address	`ip isIpv6`
`isIsoDate`	Query string returns a RFC 3339 date (`YYYY-MM-DDTHH:mm:ss.sssZ`)	`jsonpath "$.publication_date`
`isList`	Query returns a list	`jsonpath "$.books" isList`

Predicate	Description	Example
`isNumber`	Query returns an integer or a float	`jsonpath "$.count" isNumber`
`isObject`	Query returns an object (JSON object or XML node set)	`jsonpath "$.books[0]" isObje`
`isString`	Query returns a string	`jsonpath "$.name" isString`
`isUuid`	Query returns a [UUID v4]	`ip isUuid`

Each predicate can be negated by prefixing it with `not` (for instance, `not contains` or `not exists`)



A predicate value is typed, and can be a string, a boolean, a number, a bytestream, `null` or a collection. Note that `"true"` is a string, whereas `true` is a boolean.

For instance, to test the presence of a h1 node in an HTML response, the following assert can be used:

```
GET https://example.org/home
HTTP 200
[Asserts]
xpath "boolean(count(//h1))" == true
xpath "//h1" exists # Equivalent but simpler
```

As the XPath query `boolean(count(//h1))` returns a boolean, the predicate value in the assert must be either `true` or `false` without double quotes. On the other side, say you have an article node and you want to check the value of some data attributes:

```
<article
  id="electric-cars"
  data-visible="true"
...
</article>
```

The following assert will check the value of the `data-visible` attribute:

```
GET https://example.org/home
HTTP 200
[Asserts]
xpath "string(//article/@data-visible)" == "true"
```

In this case, the XPath query `string(//article/@data-visible)` returns a string, so the predicate value must be a string.

The predicate function == can be used with string, numbers or booleans; `startWith` and `contains` can only be used with strings and bytes, while `matches` only works on string. If a query returns a number, using a `matches` predicate will cause a runner error.

```
# A really well tested web page...
GET https://example.org/home
HTTP 200
[Asserts]
header "Content-Type" contains "text/html"
header "Last-Modified" == "Wed, 21 Oct 2015 07:28:00 GMT"
xpath "//h1" exists  # Check we've at least one h1
xpath "normalize-space(//h1)" contains "Welcome"
xpath "//h2" count == 13
xpath "string(//article/@data-id)" startsWith "electric"
```

Status assert

Check the received HTTP response status code. Status assert consists of the keyword `status` followed by a predicate function and value.

```
GET https://example.org
HTTP *
[Asserts]
status < 300
```

Version assert

Check the received HTTP version. Version assert consists of the keyword `version` followed by a predicate function and value. The value returns by `version` is a string:

```
GET https://example.org
HTTP *
[Asserts]
version == "2"
```

Header assert

Check the value of a received HTTP response header. Header assert consists of the keyword `header` followed by the value of the header, a predicate function and a predicate value. Like headers implicit asserts, the check is case-insensitive for the name: comparing a `Content-Type` header is equivalent to a `content-type` one.

```
GET https://example.org
HTTP 302
[Asserts]
```

```
header "Location" contains "www.example.net"
header "Last-Modified" matches /\d{2} [a-z-A-Z]{3} \d{4}/
```

If there are multiple headers with the same name, the header assert returns a collection, so `count`, `contains` can be used in this case to test the header list.

Let's say we have this request and response:

```
> GET /hello HTTP/1.1
> Host: example.org
> Accept: */*
> User-Agent: hurl/2.0.0-SNAPSHOT
>
* Response: (received 12 bytes in 11 ms)
*
< HTTP/1.0 200 OK
< Vary: Content-Type
< Vary: User-Agent
< Content-Type: text/html; charset=utf-8
< Content-Length: 12
< Server: Flask Server
< Date: Fri, 07 Oct 2022 20:53:35 GMT
```

One can use explicit header asserts:

```
GET https://example.org/hello
HTTP 200
[Asserts]
header "Vary" count == 2
header "Vary" contains "User-Agent"
header "Vary" contains "Content-Type"
```

Or implicit header asserts:

```
GET https://example.org/hello
HTTP 200
Vary: User-Agent
Vary: Content-Type
```

Cookie assert

Check value or attributes of a `Set-Cookie` response header. Cookie assert consists of the keyword `cookie`, followed by the cookie name (and optionally a cookie attribute), a predicate function and value.

Cookie attributes value can be checked by using the following format:`<cookie-name>[cookie-attribute]`. The following attributes are supported: `Value`, `Expires`, `Max-Age`, `Domain`, `Path`, `Secure`, `HttpOnly` and `SameSite`.

```
GET http://localhost:8000/cookies/set
HTTP 200

# Explicit check of Set-Cookie header value. If the attributes are
# not in this exact order, this assert will fail.
Set-Cookie: LSID=DQAAAKEaem_vYg; Expires=Wed, 13 Jan 2021 22:23:01 GMT; Secure;
HttpOnly; Path=/accounts; SameSite=Lax;
Set-Cookie: HSID=AYQEVnDKrdst; Domain=localhost; Expires=Wed, 13 Jan 2021
22:23:01 GMT; HttpOnly; Path=/
Set-Cookie: SSID=Ap4PGTEq; Domain=localhost; Expires=Wed, 13 Jan 2021 22:23:01
GMT; Secure; HttpOnly; Path=/

# Using cookie assert, one can check cookie value and various attributes.
[Asserts]
cookie "LSID" == "DQAAAKEaem_vYg"
cookie "LSID[Value]" == "DQAAAKEaem_vYg"
cookie "LSID[Expires]" exists
cookie "LSID[Expires]" contains "Wed, 13 Jan 2021"
cookie "LSID[Max-Age]" not exists
cookie "LSID[Domain]" not exists
cookie "LSID[Path]" == "/accounts"
cookie "LSID[Secure]" exists
cookie "LSID[HttpOnly]" exists
cookie "LSID[SameSite]" == "Lax"
```

> `Secure` and `HttpOnly` attributes can only be tested with `exists` or `not exists` predicates to reflect the Set-Cookie header semantics (in other words, queries `<cookie-name>[HttpOnly]` and `<cookie-name>[Secure]` don't return boolean).

Body assert

Check the value of the received HTTP response body when decoded as a string. Body assert consists of the keyword `body` followed by a predicate function and value.

```
GET https://example.org
HTTP 200
[Asserts]
body contains "<h1>Welcome!</h1>"
```

The encoding used to decode the response body bytes to a string is based on the `charset` value in the `Content-Type` header response.

```
# Our HTML response is encoded with GB 2312 (see
https://en.wikipedia.org/wiki/GB_2312)
GET https://example.org/cn
HTTP 200
[Asserts]
header "Content-Type" == "text/html; charset=gb2312"
# bytes of the response, without any text decoding:
bytes contains hex,c4e3bac3cac0bde7; # 你好世界 encoded in GB 2312
```

```
# text of the response, decoded with GB 2312:
body contains "你好世界"
```

If the `Content-Type` response header doesn't include any encoding hint, a decode filter can be used to explicitly decode the response body bytes.

```
# Our HTML response is encoded using GB 2312.
# But, the 'Content-Type' HTTP response header doesn't precise any charset,
# so we decode explicitly the bytes.
GET https://example.org/cn
HTTP 200
[Asserts]
header "Content-Type" == "text/html"
bytes contains hex,c4e3bac3cac0bde7; # 你好世界 encoded in GB2312
bytes charsetDecode "gb2312" contains "你好世界"
```

Body asserts are automatically decompressed based on the value of `Content-Encoding` response header. So, whatever is the response compression (`gzip`, `brotli`) etc... asserts values don't depend on the content encoding.

```
# Request a gzipped reponse, the `body` asserts works with ungzipped response
GET https://example.org
Accept-Encoding: gzip
HTTP 200
[Asserts]
header "Content-Encoding" == "gzip"
body contains "<h1>Welcome!</h1>"

# Without content encoding, asserts remains identical
GET https://example.org
HTTP 200
[Asserts]
header "Content-Encoding" not exists
body contains "<h1>Welcome!</h1>"
```

Bytes assert

Check the value of the received HTTP response body as a bytestream. Body assert consists of the keyword `bytes` followed by a predicate function and value.

```
GET https://example.org/data.bin
HTTP 200
[Asserts]
bytes startsWith hex,efbbbf;
bytes count == 12424
header "Content-Length" == "12424"
```

Like `body` assert, `bytes` assert works *after* content encoding decompression (so the predicates values are not affected by `Content-Encoding` response header value).

RawBytes assert

Check the value of the received HTTP response body as a raw bytestream. RawBytes assert consists of the keyword `rawbytes` followed by a predicate function and value.

```
GET https://example.org/data.bin
HTTP 200
Content-Encoding: gzip
[Asserts]
header "Content-Length" == "32"
rawbytes count == 32                # matches Content-Length (compressed size)
bytes count == 100                  # decompressed size is larger
rawbytes startsWith hex,1f8b;       # gzip magic bytes
bytes startsWith hex,48656c6c6f;    # decompressed content starts with "Hello"
```

Unlike `bytes` assert, `rawbytes` returns the raw bytes *before* any content decoding. For uncompressed responses, `rawbytes` and `bytes` return the same data.

XPath assert

Check the value of a XPath query on the received HTTP body decoded as a string (using the `charset` value in the `Content-Type` header response). Currently, only XPath 1.0 expression can be used. Body assert consists of the keyword `xpath` followed by a predicate function and value. Values can be string, boolean or number depending on the XPath query.

Let's say we want to check this HTML response:

```
$ curl -v https://example.org

< HTTP/1.1 200 OK
< Content-Type: text/html; charset=UTF-8
...
<!doctype html>
<html>
  <head>
    <title>Example Domain</title>
    ...
  </head>

  <body>
    <div>
      <h1>Example</h1>
      <p>This domain is for use in illustrative examples in documents. You may
use this domain in literature without prior coordination or asking for
permission.</p>
      <p><a href="https://www.iana.org/domains/example">More information...</a>
</p>
    </div>
  </body>
</html>
```

With Hurl, we can write multiple XPath asserts describing the DOM content:

```
GET https://example.org
HTTP 200
Content-Type: text/html; charset=UTF-8
[Asserts]
xpath "string(/html/head/title)" contains "Example" # Check title
xpath "count(//p)" == 2                             # Check the number of <p>
xpath "//p" count == 2                              # Similar assert for <p>
xpath "boolean(count(//h2))" == false               # Check there is no <h2>
xpath "//h2" not exists                             # Similar assert for <h2>
```

XML Namespaces are also supported. Let's say you want to check this XML response:

```
<?xml version="1.0"?>
<!-- both namespace prefixes are available throughout -->
<bk:book xmlns:bk='urn:loc.gov:books'
         xmlns:isbn='urn:ISBN:0-395-36341-6'>
    <bk:title>Cheaper by the Dozen</bk:title>
    <isbn:number>1568491379</isbn:number>
</bk:book>
```

This XML response can be tested with the following Hurl file:

```
GET http://localhost:8000/assert-xpath
HTTP 200
[Asserts]

xpath "string(//bk:book/bk:title)" == "Cheaper by the Dozen"
xpath "string(//*[name()='bk:book']/*[name()='bk:title'])" == "Cheaper by the
Dozen"
xpath "string(//*[local-name()='book']/*[local-name()='title'])" == "Cheaper by
the Dozen"

xpath "string(//bk:book/isbn:number)" == "1568491379"
xpath "string(//*[name()='bk:book']/*[name()='isbn:number'])" == "1568491379"
xpath "string(//*[local-name()='book']/*[local-name()='number'])" ==
"1568491379"
```

The XPath expressions `string(//bk:book/bk:title)` and `string(//bk:book/isbn:number)` are written with `bk` and `isbn` namespaces.

> For convenience, the first default namespace can be used with _

JSONPath assert

Check the value of a JSONPath query on the received HTTP body decoded as a JSON document. JSONPath assert consists of the keyword `jsonpath` followed by a predicate function and value.

Let's say we want to check this JSON response:

```
curl -v http://httpbin.org/json

< HTTP/1.1 200 OK
< Content-Type: application/json
...

{
  "slideshow": {
    "author": "Yours Truly",
    "date": "date of publication",
    "slides": [
      {
        "title": "Wake up to WonderWidgets!",
        "type": "all"
      },
       ...
    ],
    "title": "Sample Slide Show"
  }
}
```

With Hurl, we can write multiple JSONPath asserts describing the DOM content:

```
GET http://httpbin.org/json
HTTP 200
[Asserts]
jsonpath "$.slideshow.author" == "Yours Truly"
jsonpath "$.slideshow.slides[0].title" contains "Wonder"
jsonpath "$.slideshow.slides" count == 2
jsonpath "$.slideshow.date" != null
jsonpath "$.slideshow.slides[*].title" contains "Mind Blowing!"
```

> Explain that the value selected by the JSONPath is coerced to a string when only one node is selected.

In `matches` predicates, metacharacters beginning with a backslash (like `\d`, `\s`) must be escaped. Alternatively, `matches` predicate support JavaScript-like Regular expression syntax to enhance the readability:

```
GET https://example.org/hello
HTTP 200
[Asserts]

# Predicate value with matches predicate:
jsonpath "$.date" matches "^\\d{4}-\\d{2}-\\d{2}$"
jsonpath "$.name" matches "Hello [a-zA-Z]+!"

# Equivalent syntax:
```

```
jsonpath "$.date" matches /^\d{4}-\d{2}-\d{2}$/
jsonpath "$.name" matches /Hello [a-zA-Z]+!/
```

Regex assert

Check that the HTTP received body, decoded as text, matches a regex pattern.

```
GET https://example.org/hello
HTTP 200
[Asserts]
regex "^(\\d{4}-\\d{2}-\\d{2})$" == "2018-12-31"
# Same assert as previous using regex literals
regex /^(\d{4}-\d{2}-\d{2})$/ == "2018-12-31"
```

The regex pattern must have at least one capture group, otherwise the assert will fail. The assertion is done on the captured group value. When the regex pattern is a double-quoted string, metacharacters beginning with a backslash in the pattern (like `\d`, `\s`) must be escaped; literal pattern enclosed by `/` can also be used to avoid metacharacters escaping.

The regex syntax is documented at https://docs.rs/regex/latest/regex/#syntax. For instance, once can use flags to enable case-insensitive match:

```
GET https://example.org/hello
HTTP 200
[Asserts]
regex /(?i)hello (\w+)!/ == "World"
```

SHA-256 assert

Check response body SHA-256 hash.

```
GET https://example.org/data.tar.gz
HTTP 200
[Asserts]
sha256 == hex,039058c6f2c0cb492c533b0a4d14ef77cc0f78abccced5287d84a1a2011cfb81;
```

Like `body` assert, `sha256` assert works *after* content encoding decompression (so the predicates values are not affected by `Content-Encoding` response header). For instance, if we have a resource `a.txt` on a server with a given hash `abcdef`, `sha256` value is not affected by `Content-Encoding`:

```
# Without content encoding compression:
GET https://example.org/a.txt
HTTP 200
[Asserts]
sha256 == hex,abcdef;

# With content encoding compression:
```

```
GET https://example.org/a.txt
Accept-Encoding: brotli
HTTP 200
[Asserts]
header "Content-Encoding" == "brotli"
sha256 == hex,abcdef;
```

MD5 assert

Check response body MD5 hash.

```
GET https://example.org/data.tar.gz
HTTP 200
[Asserts]
md5 == hex,ed076287532e86365e841e92bfc50d8c;
```

Like `sha256` asserts, `md5` assert works *after* content encoding decompression (so the predicates values are not affected by `Content-Encoding` response header)

URL assert

Check the last fetched URL. This is most meaningful if you have told Hurl to follow redirection (see `[Options]` section(#file-format-request-options) or `--location` option). URL assert consists of the keyword `url` followed by a predicate function and value.

```
GET https://example.org/redirecting
[Options]
location: true
HTTP 200
[Asserts]
url == "https://example.org/redirected"
```

Redirects assert

Check each step of redirection. This is most meaningful if you have told Hurl to follow redirection (see `[Options]` section(#file-format-request-options) or `--location` option). Redirects assert consists of the keyword `redirects` followed by a predicate function and value. The `redirects` query returns a collection of redirections that can be tested with a `location` filter:

```
GET https://example.org/redirecting/1
[Options]
location: true
HTTP 200
[Asserts]
redirects count == 3
redirects nth 0 location == "https://example.org/redirecting/2"
redirects nth 1 location == "https://example.org/redirecting/3"
redirects nth 2 location == "https://example.org/redirected"
```

IP address assert

Check the IP address of the last connection. The value of the `ip` query is a string.

> Predicates `isIpv4` and `isIpv6` are available to check if a particular string matches an IPv4 or IPv6 address and can use with `ip` queries.

```
GET https://example.org/hello
HTTP 200
[Asserts]
ip isIpv4
ip not isIpv6
ip == "172.16.45.87"
```

Variable assert

```
# Test that the XML endpoint return 200 pets
GET https://example.org/api/pets
HTTP 200
[Captures]
pets: xpath "//pets"
[Asserts]
variable "pets" count == 200
```

Duration assert

Check the total duration (sending plus receiving time) of the HTTP transaction.

```
GET https://example.org/helloworld
HTTP 200
[Asserts]
duration < 1000   # Check that response time is less than one second
```

SSL certificate assert

Check the SSL certificate properties. Certificate assert consists of the keyword `certificate`, followed by the certificate attribute value.

The following attributes are supported: `Subject`, `Issuer`, `Start-Date`, `Expire-Date`, `Serial-Number`, `Subject-Alt-Name` and `Value`.

```
GET https://example.org
HTTP 200
[Asserts]
certificate "Subject" == "CN=example.org"
certificate "Issuer" == "C=US, O=Let's Encrypt, CN=R3"
certificate "Expire-Date" daysAfterNow > 15
certificate "Serial-Number" matches "[0-9af]+"
```

```
certificate "Subject-Alt-Name" contains "DNS:example.org"
certificate "Subject-Alt-Name" split "," count == 2
certificate "Value" startsWith "-----BEGIN CERTIFICATE-----"
```

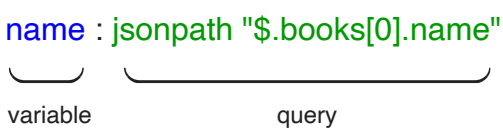
Filters

Definition

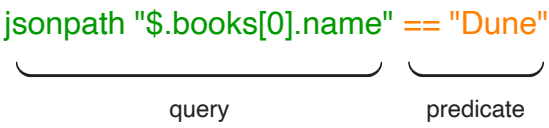
Captures and asserts share a common structure: query. A query is used to extract data from an HTTP response; this data can come from the HTTP response body, the HTTP response headers or from the HTTP meta-information (like `duration` for instance)…

In this example, the query **`jsonpath "$.books[0].name"`** is used in a capture to save data and in an assert to test the HTTP response body.

Capture:

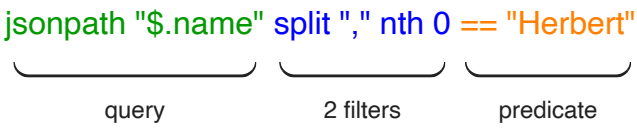


Assert:



In both case, the query is exactly the same: queries are the core structure of asserts and captures. Sometimes, you want to process data extracted by queries: that's the purpose of **filters**.

Filters are used to transform value extracted by a query and can be used in asserts and captures to refine data. Filters **can be chained**, allowing for fine-grained data extraction.



Example

```
GET https://example.org/api
HTTP 200
[Captures]
name: jsonpath "$.user.id" replaceRegex /\d/ "x"
```

```
[Asserts]
header "x-servers" split "," count == 2
header "x-servers" split "," nth 0 == "rec1"
header "x-servers" split "," nth 1 == "rec3"
jsonpath "$.books" count == 12
```

Description

Filter	Description	Input	Outpu
base64Decode	Decodes a [Base64 encoded string] into bytes.	string	bytes
base64Encode	Encodes bytes into [Base64 encoded string].	bytes	string
base64UrlSafeDecode	Decodes a Base64 encoded string into bytes (using [Base64 URL safe encoding]).	string	bytes
base64UrlSafeEncode	Encodes bytes into Base64 encoded string (using [Base64 URL safe encoding]).	bytes	string
charsetDecode	Decodes bytes to string using a charset encoding.	bytes	string
count	Counts the number of items in a collection.	collection	numbe
dateFormat	Formats a date to a string given [a specification format].	date	string
daysAfterNow	Returns the number of days between now and a date in the future.	date	numbe
daysBeforeNow	Returns the number of days between now and a date in the past.	date	numbe
first	Returns the first element from a collection.	collection	any
htmlEscape	Converts the characters &, < and > to HTML-safe sequence.	string	string
htmlUnescape	Converts all named and numeric character references (e.g. >, >, >) to the corresponding Unicode characters.	string	string
jsonpath	Evaluates a JSONPath expression.	string	any
last	Returns the last element from a collection.	collection	any
location	Returns the target location URL of a redirection.	response	string

Filter	Description	Input	Outpu
nth	Returns the element from a collection at a zero-based index, accepts negative indices for indexing from the end of the collection.	collectio	any
regex	Extracts regex capture group. Pattern must have at least one capture group.	string	string
replace	Replaces all occurrences of old string with new string.	string	string
replaceRegex	Replaces all occurrences of a pattern with new string.	string	string
split	Splits to a list of strings around occurrences of the specified delimiter.	string	string
toDate	Converts a string to a date given [a specification format].	string	date
toFloat	Converts value to float number.	string \	numbe
toHex	Converts bytes to hexadecimal string.	bytes	string
toInt	Converts value to integer number.	string \	numbe
toString	Converts value to string.	any	string
urlDecode	Replaces %xx escapes with their single-character equivalent.	string	string
urlEncode	Percent-encodes all the characters which are not included in unreserved chars (see [RFC3986]) with the exception of forward slash (/).	string	string
urlQueryParam	Returns the value of a query parameter in a URL.	string	string
utf8Decode	Decodes bytes to string using UTF-8 encoding.	bytes	string
utf8Encode	Encodes a string to bytes using UTF-8 encoding.	string	bytes
xpath	Evaluates a XPath expression.	string	string

base64Decode

Decodes a Base64 encoded string into bytes.

```
GET https://example.org/api
HTTP 200
[Asserts]
jsonpath "$.token" base64Decode == hex,3c3c3f3f3f3e3e;
```

base64Encode

Encodes bytes into Base64 encoded string.

```
GET https://example.org/api
HTTP 200
[Asserts]
bytes base64Encode == "PDw/Pz8+Pg=="
```

base64UrlSafeDecode

Decodes a Base64 encoded string into bytes (using Base64 URL safe encoding).

```
GET https://example.org/api
HTTP 200
[Asserts]
jsonpath "$.token" base64UrlSafeDecode == hex,3c3c3f3f3f3e3e;
```

base64UrlSafeEncode

Encodes bytes into Base64 encoded string (using Base64 URL safe encoding).

```
GET https://example.org/api
HTTP 200
[Asserts]
bytes base64UrlSafeEncode == "PDw_Pz8-Pg"
```

charsetDecode

Formerly known as `decode`, which is deprecated and will be removed in a future major version.

Decodes bytes to string using a charset encoding. Encoding labels are defined in Encoding Standard.

```
# The 'Content-Type' HTTP response header does not precise the charset 'gb2312'
# so body must be decoded explicitly by Hurl before processing any text based
assert
GET https://example.org/hello_china
HTTP 200
[Asserts]
header "Content-Type" == "text/html"
# Content-Type has no encoding clue, we must decode ourselves the body
```

```
response.
bytes charsetDecode "gb2312" xpath "string(//body)" == "你好世界"
```

When the encoding is UTF-8 (i.e. `charsetDecode "utf-8"`), an `utf8Decode` filter can be used instead.

count

Counts the number of items in a collection.

```
GET https://example.org/api
HTTP 200
[Asserts]
jsonpath "$.books" count == 12
```

dateFormat

Formerly known as `format`, which is deprecated and will be removed in a future major version.

Formats a date to a string given a specification format.

```
GET https://example.org
HTTP 200
[Asserts]
cookie "LSID[Expires]" dateFormat "%a, %d %b %Y %H:%M:%S" == "Wed, 13 Jan 2021
22:23:01"
```

daysAfterNow

Returns the number of days between now and a date in the future.

```
GET https://example.org
HTTP 200
[Asserts]
certificate "Expire-Date" daysAfterNow > 15
```

daysBeforeNow

Returns the number of days between now and a date in the past.

```
GET https://example.org
HTTP 200
[Asserts]
certificate "Start-Date" daysBeforeNow < 100
```

first

Returns the first element from a collection.

```
GET https://example.org
HTTP 200
[Asserts]
jsonpath "$.books" first == "Dune"
```

htmlEscape

Converts the characters &, < and > to HTML-safe sequence.

```
GET https://example.org/api
HTTP 200
[Asserts]
jsonpath "$.text" htmlEscape == "a &gt; b"
```

htmlUnescape

Converts all named and numeric character references (e.g. `>`, `>`, `>`) to the corresponding Unicode characters.

```
GET https://example.org/api
HTTP 200
[Asserts]
jsonpath "$.escaped_html[1]" htmlUnescape == "Foo © bar ≧"
```

jsonpath

Evaluates a JSONPath expression.

```
GET https://example.org/api
HTTP 200
[Captures]
books: xpath "string(//body/@data-books)"
[Asserts]
variable "books" jsonpath "$[0].name" == "Dune"
variable "books" jsonpath "$[0].author" == "Franck Herbert"
```

last

Returns the last element from a collection.

```
GET https://example.org
HTTP 200
[Asserts]
jsonpath "$.books" last == "Les Misérables"
```

location

Returns the target URL location of a redirection; the returned URL is always absolute, contrary to the `Location` header from which it's originated that can be absolute or relative.

```
GET https://example.org/step1
[Options]
location: true
HTTP 200
[Asserts]
redirects count == 2
redirects nth 0 location == "https://example.org/step2"
redirects nth 1 location == "https://example.org/step3"
```

nth

Returns the element from a collection at a zero-based index, accepts negative indices for indexing from the end of the collection.

```
GET https://example.org/api
HTTP 200
[Asserts]
jsonpath "$.books" nth 2 == "Children of Dune"
```

regex

Extracts regex capture group. Pattern must have at least one capture group.

```
GET https://example.org/foo
HTTP 200
[Captures]
param1: header "header1"
param2: header "header2" regex "Hello (.*)!"
param3: header "header2" regex /Hello (.*)!/
param3: header "header2" regex /(?i)Hello (.*)!/
```

The regex syntax is documented at https://docs.rs/regex/latest/regex/#syntax.

replace

Replaces all occurrences of old string with new string.

```
GET https://example.org/foo
HTTP 200
[Captures]
url: jsonpath "$.url" replace "http://" "https://"
[Asserts]
jsonpath "$.ips" replace ", " "|" == "192.168.2.1|10.0.0.20|10.0.0.10"
```

replaceRegex

Replaces all occurrences of a pattern with new string.

```
GET https://example.org/foo
HTTP 200
[Captures]
url: jsonpath "$.id" replaceRegex /\d/ "x"
[Asserts]
jsonpath "$.message" replaceRegex "B[aoi]b" "Dude" == "Welcome Dude!"
```

split

Splits to a list of strings around occurrences of the specified delimiter.

```
GET https://example.org/foo
HTTP 200
[Asserts]
jsonpath "$.ips" split ", " count == 3
```

toDate

Converts a string to a date given a specification format.

```
GET https:///example.org
HTTP 200
[Asserts]
header "Expires" toDate "%a, %d %b %Y %H:%M:%S GMT" daysBeforeNow > 1000
```

ISO 8601 / RFC 3339 date and time format have shorthand format `%+`:

```
GET https://example.org/api/books
HTTP 200
[Asserts]
jsonpath "$.published" == "2023-01-23T18:25:43.511Z"
jsonpath "$.published" toDate "%Y-%m-%dT%H:%M:%S%.fZ" dateFormat "%A" ==
"Monday"
jsonpath "$.published" toDate "%+" dateFormat "%A" == "Monday" # %+ can be used
to parse ISO 8601 / RFC 3339
```

toFloat

Converts value to float number.

```
GET https://example.org/foo
HTTP 200
```

```
[Asserts]
jsonpath "$.pi" toFloat == 3.14
```

toHex

Converts bytes to hexadecimal string.

```
GET https://example.org/foo
HTTP 200
[Asserts]
bytes toHex == "d188d0b5d0bbd0bbd18b"
```

toInt

Converts value to integer number.

```
GET https://example.org/foo
HTTP 200
[Asserts]
jsonpath "$.id" toInt == 123
```

toString

Converts value to string.

```
GET https://example.org/foo
HTTP 200
[Asserts]
jsonpath "$.count" toString == "42"
```

urlDecode

Replaces %xx escapes with their single-character equivalent.

```
GET https://example.org/foo
HTTP 200
[Asserts]
jsonpath "$.encoded_url" urlDecode == "https://mozilla.org/?x=шеллы"
```

urlEncode

Percent-encodes all the characters which are not included in unreserved chars (see RFC3986) except forward slash (/).

```
GET https://example.org/foo
HTTP 200
[Asserts]
```

```
jsonpath "$.url" urlEncode ==
"https%3A//mozilla.org/%3Fx%3D%D1%88%D0%B5%D0%BB%D0%BB%D1%8B"
```

urlQueryParam

Returns the value of a query parameter in a URL.

```
GET https://example.org/foo
HTTP 200
[Asserts]
jsonpath "$.url" urlQueryParam "x" == "шеллы"
```

utf8Decode

Decodes bytes to string using UTF-8 encoding.

```
GET https://example.org/messages
HTTP 200
[Asserts]
# From a Base64 string to UTF-8 bytes to final string
jsonpath "$.bytesInBase64" base64Decode utf8Decode == "Hello World"
```

utf8Encode

Encodes a string to bytes using UTF-8 encoding.

```
GET https://example.org/drinks
HTTP 200
[Asserts]
jsonpath "$.beverage" utf8Encode toHex == "636166C3A9"
```

xpath

Evaluates a XPath expression.

```
GET https://example.org/hello_gb2312
HTTP 200
[Asserts]
bytes charsetDecode "gb2312" xpath "string(//body)" == "你好世界"
```

Templates

Variables

In Hurl file, you can generate value using two curly braces, i.e `{{my_variable}}`. For instance, if you want to reuse a value from an HTTP response in the next entries, you can capture this value in a variable and reuse it in a placeholder.

In this example, we capture the value of a CSRF token from the body of the first response, and inject it as a header in the next POST request:

```
GET https://example.org
HTTP 200
[Captures]
csrf_token: xpath "string(//meta[@name='_csrf_token']/@content)"

# Do the login !
POST https://acmecorp.net/login?user=toto&password=1234
X-CSRF-TOKEN: {{csrf_token}}
HTTP 302
```

In this second example, we capture the body in a variable `index`, and reuse this value in the query `jsonpath "$.errors[{{index}}].id"`:

```
GET https://example.org/api/index
HTTP 200
[Captures]
index: body

GET https://example.org/api/status
HTTP 200
[Asserts]
jsonpath "$.errors[{{index}}].id" == "error"
```

Functions

Besides variables, functions can be used to generate dynamic values. Current functions are:

Function	Description
`newUuid`	Generates an UUID v4 random string
`newDate`	Generates an RFC 3339 UTC date string, at the current time

In the following example, we use `newDate` to generate a dynamic query parameter:

```
GET https://example.org/api/foo
[Query]
date: {{newDate}}
HTTP 200
```

We run a `GET` request to `https://example.org/api/foo?date=2024%2D12%2D02T10%3A35%3A44%2E461731Z` where the `date` query parameter value is `2024-12-02T10:35:44.461731Z` URL encoded.

In this second example, we use `newUuid` function to generate an email dynamically:

```
POST https://example.org/api/foo
{
  "name": "foo",
  "email": "{{newUuid}}@test.com"
}
```

When run, the request body will be:

```
{
  "name": "foo",
  "email": "0531f78f-7f87-44be-a7f2-969a1c4e6d97@test.com"
}
```

Types

Values generated from function and variables are typed, and can be either string, bool, number, `null` or collections. Depending on the value type, templates can be rendered differently. Let's say we have captured an integer value into a variable named `count`:

```
GET https://sample/counter

HTTP 200
[Captures]
count: jsonpath "$.results[0]"
```

The following entry:

```
GET https://sample/counter/{{count}}

HTTP 200
[Asserts]
jsonpath "$.id" == "{{count}}"
```

will be rendered at runtime to:

```
GET https://sample/counter/458

HTTP 200
[Asserts]
jsonpath "$.id" == "458"
```

resulting in a comparison between the JSONPath expression and a string value.

On the other hand, the following assert:

```
GET https://sample/counter/{{count}}

HTTP 200
[Asserts]
jsonpath "$.index" == {{count}}
```

will be rendered at runtime to:

```
GET https://sample/counter/458

HTTP 200
[Asserts]
jsonpath "$.index" == 458
```

resulting in a comparison between the JSONPath expression and an integer value.

So if you want to use typed values (in asserts for instances), you can use `{{my_var}}`. If you're interested in the string representation of a variable, you can surround the variable with double quotes , as in `"{{my_var}}"`.

> When there is no possible ambiguities, like using a variable in an URL, or in a header, you can omit the double quotes. The value will always be rendered as a string.

Injecting Variables

Variables can be injected in a Hurl file:

- by using `--variable` option
- by using `--variables-file` option
- by defining environment variables, for instance `HURL_VARIABLE_foo=bar`
- by defining variables in an `[Options]` section(#file-format-request-options)
- by defining variables in Hurl configuration file

Lets' see how to inject variables, given this `test.hurl`:

```
GET https://{{host}}/{{id}}/status
HTTP 304
```

```
GET https://{{host}}/health
HTTP 200
```

`variable` option

Variable can be defined with command line option:

```
$ hurl --variable host=example.net --variable id=1234 test.hurl
```

`variables-file` option

We can also define all injected variables in a file:

```
$ hurl --variables-file vars.env test.hurl
```

where `vars.env` is

```
host=example.net
id=1234
```

Environment variable

We can use environment variables in the form of `HURL_VARIABLE_name=value`:

```
$ export HURL_VARIABLE_host=example.net
$ export HURL_VARIABLE_id=1234
$ hurl test.hurl
```

Options sections

We can define variables in `[Options]` section. Variables defined in a section are available for the next requests.

```
GET https://{{host}}/{{id}}/status
[Options]
variable: host=example.net
variable: id=1234
HTTP 304

GET https://{{host}}/health
HTTP 200
```

Secrets

Secrets are variables which value is redacted from standard error logs (for instance using `--very-verbose`) and reports. Secrets are injected through command-line with `--secret` option:

```
$ hurl --secret token=FooBar test.hurl
```

Values are redacted by *exact matching*: if a secret value is transformed, and you want to redact also the transformed value, you can add as many secrets as there are transformed values. Even if a secret is not used as a variable, all secrets values will be redacted from messages and logs.

```
$ hurl --secret token=FooBar \
       --secret token_alt_0=FOOBAR \
       --secret token_alt_1=foobar \
       test.hurl
```

> Secrets **are not redacted** from HTTP responses outputted on standard output as Hurl considers the standard output as the correct unaltered output of a run. With this call `$ hurl --secret token=FooBar test.hurl`, the HTTP response is outputted unaltered and `FooBar` can appear in the HTTP response. Options that transforms Hurl output on standard output, like `--include` or `--json` works the same. JSON report also saves each unaltered HTTP response on disk so extra care must be taken when secrets are in the HTTP response body.

Templating Body

Variables and functions can be used in JSON body:

```
PUT https://example.org/api/hits
{
    "key0": "{{a_string}}",
    "key1": {{a_bool}},
    "key2": {{a_null}},
    "key3": {{a_number}},
    "key4": "{{newDate}}"
}
```

Note that we're writing a kind of JSON body directly without any delimitation marker. For the moment, XML body can't use variables directly. In order to templatize a XML body, you can use multiline string body with variables and functions. The multiline string body allows to templatize any text based body (JSON, XML, CSV etc…):

Multiline string body delimited by ```:

````
PUT https://example.org/api/hits
Content-Type: application/json
```
{
    "key0": "{{a_string}}",
    "key1": {{a_bool}},
    "key2": {{a_null}},
    "key3": {{a_number}},
    "key4: "{{newDate}}"
````
````

```
}
```
```

Variables can be initialized via command line:

```
$ hurl --variable a_string=apple --variable a_bool=true --variable a_null=null
--variable a_number=42 test.hurl
```

Resulting in a PUT request with the following JSON body:

```
{
    "key0": "apple",
    "key1": true,
    "key2": null,
    "key3": 42,
    "key4": "2024-12-02T13:39:45.936643Z"
}
```
```

Grammar

Definitions

Short description:

- operator | denotes alternative,
- operator * denotes iteration (zero or more),
- operator + denotes iteration (one or more),

Syntax Grammar

General

```
hurl_file
  entry*
  lt*

entry(used by hurl_file)
  request
  response?

request(used by entry)
  lt*
  method sp value_string lt
  header*
  request_section*
  body?

response(used by entry)
  lt*
  version sp status lt
  header*
  response_section*
  body?

method(used by request)
  [A-Z]+

version(used by response)
   HTTP/1.0
  |HTTP/1.1
  |HTTP/2
  |HTTP

status(used by response)
```

```
  [0-9]+

header(used by request, response)
  lt*
  key-value lt

body(used by request, response)
  lt*
  bytes lt
```

Sections

```
request-section(used by request)
   basic-auth-section
  |query-string-params-section
  |form-params-section
  |multipart-form-data-section
  |cookies-section
  |options-section

response-section(used by response)
   captures-section
  |asserts-section

query-string-params-section(used by request-section)
  lt*
  ([QueryStringParams]|[Query]) lt
  key-value*

form-params-section(used by request-section)
  lt*
  ([FormParams]|[Form]) lt
  key-value*

multipart-form-data-section(used by request-section)
  lt*
  ([MultipartFormData]|[Multipart]) lt
  multipart-form-data-param*

cookies-section(used by request-section)
  lt*
  [Cookies] lt
  key-value*

captures-section(used by response-section)
  lt*
  [Captures] lt
  capture*
```

```
asserts-section(used by response-section)
  lt*
  [Asserts] lt
  assert*

basic-auth-section(used by request-section)
  lt*
  [BasicAuth] lt
  key-value*

options-section(used by request-section)
  lt*
  [Options] lt
  option*

key-value(used by header, query-string-params-section, form-params-
section, cookies-section, basic-auth-section, multipart-form-data-param)
  key-string : value-string

multipart-form-data-param(used by multipart-form-data-section)
  filename-param|key-value

filename-param(used by multipart-form-data-param)
  lt*
  key-string : filename-value lt

filename-value(used by filename-param)
  file, filename ; (filename-content-type)?

filename-content-type(used by filename-value)
  value-string

capture(used by captures-section)
  lt*
  key-string : query (sp filter)* (sp redact)? lt

assert(used by asserts-section)
  lt*
  query (sp filter)* sp predicate lt

option(used by options-section)
  lt*
  (aws-sigv4-option|ca-certificate-option|client-certificate-
  option|client-key-option|compressed-option|connect-to-option|connect-
  timeout-option|delay-option|digest-option|fail-with-body-option|follow-
  redirect-option|follow-redirect-trusted-option|header-option|http10-
  option|http11-option|http2-option|http3-option|insecure-option|ipv4-
  option|ipv6-option|limit-rate-option|max-redirs-option|max-time-
  option|negotiate-option|netrc-option|netrc-file-option|netrc-optional-
  option|output-option|path-as-is-option|pinned-public-key-option|proxy-
```

option|repeat-option|resolve-option|retry-option|retry-interval-option|skip-option|unix-socket-option|user-option|variable-option|variables-file-option|verbose-option|verbosity-option|very-verbose-option)

aws-sigv4-option(used by option)
  aws-sigv4 : value-string lt

ca-certificate-option(used by option)
  cacert : filename lt

client-certificate-option(used by option)
  cert : filename-password lt

client-key-option(used by option)
  key : value-string lt

compressed-option(used by option)
  compressed : boolean-option lt

connect-to-option(used by option)
  connect-to : value-string lt

connect-timeout-option(used by option)
  connect-timeout : duration-option lt

delay-option(used by option)
  delay : duration-option lt

digest-option(used by option)
  digest : boolean-option lt

fail-with-body-option(used by option)
  fail-with-body : boolean-option lt

follow-redirect-option(used by option)
  location : boolean-option lt

follow-redirect-trusted-option(used by option)
  location-trusted : boolean-option lt

header-option(used by option)
  header : value-string lt

http10-option(used by option)
  http1.0 : boolean-option lt

http11-option(used by option)
  http1.1 : boolean-option lt

http2-option(used by option)

```
  http2 : boolean-option lt
```

http3-option(used by option)
```
  http3 : boolean-option lt
```

insecure-option(used by option)
```
  insecure : boolean-option lt
```

ipv4-option(used by option)
```
  ipv4 : boolean-option lt
```

ipv6-option(used by option)
```
  ipv6 : boolean-option lt
```

limit-rate-option(used by option)
```
  limit-rate : integer-option lt
```

max-redirs-option(used by option)
```
  max-redirs : integer-option lt
```

max-time-option(used by option)
```
  max-time : integer-option lt
```

negotiate-option(used by option)
```
  negotiate : boolean-option lt
```

netrc-option(used by option)
```
  netrc : boolean-option lt
```

netrc-file-option(used by option)
```
  netrc-file : value-string lt
```

netrc-optional-option(used by option)
```
  netrc-optional : boolean-option lt
```

output-option(used by option)
```
  output : value-string lt
```

path-as-is-option(used by option)
```
  path-as-is : boolean-option lt
```

pinned-public-key-option(used by option)
```
  pinnedpubkey : value-string lt
```

proxy-option(used by option)
```
  proxy : value-string lt
```

resolve-option(used by option)
```
  resolve : value-string lt
```

repeat-option(used by option)

```
  repeat : integer-option lt

retry-option(used by option)
  retry : integer-option lt

retry-interval-option(used by option)
  retry-interval : duration-option lt

skip-option(used by option)
  skip : boolean-option lt

unix-socket-option(used by option)
  unix-socket : value-string lt

user-option(used by option)
  user : value-string lt

variable-option(used by option)
  variable : variable-definition lt

variables-file-option(used by option)
  variables-file : filename lt

verbose-option(used by option)
  verbose : boolean-option lt

verbosity-option(used by option)
  verbosity : (brief|verbose|debug)

very-verbose-option(used by option)
  very-verbose : boolean-option lt

variable-definition(used by variable-option)
  variable-name = variable-value

boolean-option(used by compressed-option, digest-option, fail-with-body-
option, follow-redirect-option, follow-redirect-trusted-option, http10-
option, http11-option, http2-option, http3-option, insecure-option, ipv4-
option, ipv6-option, negotiate-option, netrc-option, netrc-optional-option, path-
as-is-option, skip-option, verbose-option, very-verbose-option)
  boolean|placeholder

integer-option(used by limit-rate-option, max-redirs-option, max-time-
option, repeat-option, retry-option)
  integer|placeholder

duration-option(used by connect-timeout-option, delay-option, retry-interval-
option)
  (integer duration-unit?)|placeholder

duration-unit(used by duration-option)
```

```
  ms|s|m

variable-value(used by variable-definition)
   null
  |boolean
  |integer
  |float
  |key-string
  |quoted-string
```

Query

```
query(used by capture, assert)
   status-query
  |version-query
  |url-query
  |ip-query
  |header-query
  |certificate-query
  |cookie-query
  |body-query
  |xpath-query
  |jsonpath-query
  |regex-query
  |variable-query
  |duration-query
  |bytes-query
  |sha256-query
  |md5-query

status-query(used by query)
  status

version-query(used by query)
  version

url-query(used by query)
  url

ip-query(used by query)
  ip

header-query(used by query)
  header sp quoted-string

certificate-query(used by query)
  certificate sp (Subject|Issuer|Start-Date|Expire-Date|Serial-Number)

cookie-query(used by query)
```

```
  cookie sp quoted-string

body-query(used by query)
  body

xpath-query(used by query)
  xpath sp quoted-string

jsonpath-query(used by query)
  jsonpath sp quoted-string

regex-query(used by query)
  regex sp (quoted-string|regex)

variable-query(used by query)
  variable sp quoted-string

duration-query(used by query)
  duration

sha256-query(used by query)
  sha256

md5-query(used by query)
  md5

bytes-query(used by query)
  bytes
```

Predicates

```
predicate(used by assert)
  (not sp)? predicate-func

predicate-func(used by predicate)
   equal-predicate
  |boolean-predicate
  |contain-predicate
  |end-with-predicate
  |exist-predicate
  |greater-or-equal-predicate
  |greater-predicate
  |include-predicate
  |is-collection-predicate
  |is-date-predicate
  |is-float-predicate
  |is-empty-predicate
  |is-integer-predicate
  |is-ipv4-predicate
```

```
  |is-ipv6-predicate
  |is-iso-date-predicate
  |is-list
  |is-object
  |is-string-predicate
  |is-uuid-predicate
  |less-or-equal-predicate
  |less-predicate
  |match-predicate
  |not-equal-predicate
  |start-with-predicate

equal-predicate(used by predicate-func)
  == sp predicate-value

boolean-predicate(used by predicate-func)
  isBoolean

contain-predicate(used by predicate-func)
  contains sp quoted-string

end-with-predicate(used by predicate-func)
  endsWith sp (quoted-string|oneline-hex|oneline-base64)

exist-predicate(used by predicate-func)
  exists

greater-or-equal-predicate(used by predicate-func)
  >= sp (number|quoted-string|placeholder)

greater-predicate(used by predicate-func)
  > sp (number|quoted-string|placeholder)

include-predicate(used by predicate-func)
  includes sp predicate-value

is-collection-predicate(used by predicate-func)
  isCollection

is-date-predicate(used by predicate-func)
  isDate

is-empty-predicate(used by predicate-func)
  isEmpty

is-float-predicate(used by predicate-func)
  isFloat

is-integer-predicate(used by predicate-func)
  isInteger
```

is-ipv4-predicate(used by predicate-func)
  isIpv4

is-ipv6-predicate(used by predicate-func)
  isIpv6

is-iso-date-predicate(used by predicate-func)
  isIsoDate

is-list(used by predicate-func)
  isList

is-object(used by predicate-func)
  isObject

is-string-predicate(used by predicate-func)
  isString

is-uuid-predicate(used by predicate-func)
  isUuid

less-or-equal-predicate(used by predicate-func)
  <= sp (number|quoted-string|placeholder)

less-predicate(used by predicate-func)
  < sp (number|quoted-string|placeholder)

match-predicate(used by predicate-func)
  matches sp (quoted-string|regex)

not-equal-predicate(used by predicate-func)
  != sp predicate-value

start-with-predicate(used by predicate-func)
  startsWith sp (quoted-string|oneline-hex|oneline-base64)

predicate-value(used by equal-predicate, include-predicate, not-equal-predicate)
   boolean
  |multiline-string
  |null
  |number
  |oneline-string
  |oneline-base64
  |oneline-file
  |oneline-hex
  |quoted-string
  |placeholder

Bytes

bytes(used by body)
```
   json-value
  |xml
  |multiline-string
  |oneline-string
  |oneline-base64
  |oneline-file
  |oneline-hex
```

xml(used by bytes)
```
  < To Be Defined >
```

oneline-base64(used by end-with-predicate, start-with-predicate, predicate-value, bytes)
```
  base64, [A-Z0-9+-= \n]+ ;
```

oneline-file(used by predicate-value, bytes)
```
  file, filename ;
```

oneline-hex(used by end-with-predicate, start-with-predicate, predicate-value, bytes)
```
  hex, hexdigit* ;
```

Strings

quoted-string(used by variable-value, header-query, cookie-query, xpath-query, jsonpath-query, regex-query, variable-query, contain-predicate, end-with-predicate, greater-or-equal-predicate, greater-predicate, less-or-equal-predicate, less-predicate, match-predicate, start-with-predicate, predicate-value, charset-decode-filter, date-format-filter, jsonpath-filter, regex-filter, replace-filter, replace-regex-filter, split-filter, to-date-filter, url-query-param-filter, xpath-filter)
```
  " (quoted-string-content|placeholder)* "
```

quoted-string-content(used by quoted-string)
```
  (quoted-string-text|quoted-string-escaped-char)*
```

quoted-string-text(used by quoted-string-content)
```
  ~["\\]+
```

quoted-string-escaped-char(used by quoted-string-content)
```
  \ ("|\|\b|\f|\n|\r|\t|\u unicode-char)
```

key-string(used by key-value, filename-param, capture, variable-value)
```
  (key-string-content|placeholder)+
```

key-string-content(used by key-string)
```
  (key-string-text|key-string-escaped-char)*
```

```
key-string-text(used by key-string-content)
  (alphanum|_|-|.|[|]|@|$)+

key-string-escaped-char(used by key-string-content)
  \ (#|:|\|\b|\f|\n|\r|\t|\u unicode-char)

value-string(used by request, key-value, filename-content-type, aws-sigv4-
option, client-key-option, connect-to-option, header-option, netrc-file-
option, output-option, pinned-public-key-option, proxy-option, resolve-
option, unix-socket-option, user-option)
  (value-string-content|placeholder)*

value-string-content(used by value-string)
  (value-string-text|value-string-escaped-char)*

value-string-text(used by value-string-content)
  ~[#\n\\]+

value-string-escaped-char(used by value-string-content)
  \ (#|\|\b|\f|\n|\r|\t|\u unicode-char)

oneline-string(used by predicate-value, bytes)
  ` (oneline-string-content|placeholder)* `

oneline-string-content(used by oneline-string)
  (oneline-string-text|oneline-string-escaped-char)*

oneline-string-text(used by oneline-string-content)
  ~[#\n\\] ~`

oneline-string-escaped-char(used by oneline-string-content)
  \ (`|#|\|b|f|u unicode-char)

multiline-string(used by predicate-value, bytes)
  ``` multiline-string-type? lt
  (multiline-string-content|placeholder)* lt
  ```

multiline-string-type(used by multiline-string)
   base64
  |hex
  |json
  |xml
  |graphql
  |raw

multiline-string-content(used by multiline-string)
  (multiline-string-text|multiline-string-escaped-char)*

multiline-string-text(used by multiline-string-content)
```

```
  ~[\\]+ ~```

multiline-string-escaped-char(used by multiline-string-content)
  \ (\|b|f|n|r|t|`|u unicode-char)

filename(used by filename-value, ca-certificate-option, variables-file-option, oneline-file)
  (filename-content|placeholder)*

filename-content(used by filename)
  (filename-text|filename-escaped-char)*

filename-text(used by filename-content)
  ~[#;{} \n\r\\]+

filename-escaped-char(used by filename-content)
  \ (\|b|f|n|r|t|#|;| |{|}|u unicode-char)

filename-password(used by client-certificate-option)
  (filename-password-content|placeholder)*

filename-password-content(used by filename-password)
  (filename-password-text|filename-password-escaped-char)*

filename-password-text(used by filename-password-content)
  ~[#;{} \n\r\\]+

filename-password-escaped-char(used by filename-password-content)
  \ (\|b|f|n|r|t|#|;| |{|}|:|u unicode-char)

unicode-char(used by quoted-string-escaped-char, key-string-escaped-char, value-string-escaped-char, oneline-string-escaped-char, multiline-string-escaped-char, filename-escaped-char, filename-password-escaped-char)
  { hexdigit+ }
```

JSON

```
json-value(used by bytes, json-key-value, json-array)
   placeholder
  |json-object
  |json-array
  |json-string
  |json-number
  |boolean
  |null

json-object(used by json-value)
  { json-key-value (, json-key-value)* }

json-key-value(used by json-object)
```

```
  json-string : json-value

json-array(used by json-value)
  [ json-value (, json-value)* ]

json-string(used by json-value, json-key-value)
  " (json-string-content|placeholder)* "

json-string-content(used by json-string)
  json-string-text|json-string-escaped-char

json-string-text(used by json-string-content)
  ~["\\]

json-string-escaped-char(used by json-string-content)
  \ ("|\|b|f|n|r|t|u hexdigit hexdigit hexdigit hexdigit)

json-number(used by json-value)
  [-]? json-integer fraction? exponent?

json-integer(used by json-number)
  0|[1-9] digit*
```

Expression

```
placeholder(used by boolean-option, integer-option, duration-option, greater-or-equal-predicate, greater-predicate, less-or-equal-predicate, less-predicate, predicate-value, quoted-string, key-string, value-string, oneline-string, multiline-string, filename, filename-password, json-value, json-string, nth-filter)
  {{ expr }}

expr(used by placeholder)
  (variable-name|function) (sp filter)*

variable-name(used by variable-definition, expr)
  [A-Za-z] [A-Za-z_-0-9]*
```

Function

```
function(used by expr)
   env-function
  |now-function
  |uuid-function

env-function(used by function)
  getEnv

now-function(used by function)
```

newDate

uuid-function(used by function)
newUuid

Filter

filter(used by capture, assert, expr)
base64-decode-filter
|base64-encode-filter
|base64-url-safe-decode-filter
|base64-url-safe-encode-filter
|charset-decode-filter
|count-filter
|days-after-now-filter
|days-before-now-filter
|first-filter
|date-format-filter
|html-escape-filter
|html-unescape-filter
|jsonpath-filter
|last-filter
|location-filter
|nth-filter
|regex-filter
|replace-filter
|replace-regex-filter
|split-filter
|to-date-filter
|to-float-filter
|to-hex-filter
|to-int-filter
|to-string-filter
|url-decode-filter
|url-encode-filter
|url-query-param-filter
|utf8-decode-filter
|utf8-encode-filter
|xpath-filter

base64-decode-filter(used by filter)
base64Decode

base64-encode-filter(used by filter)
base64Encode

base64-url-safe-decode-filter(used by filter)
base64UrlSafeDecode

base64-url-safe-encode-filter(used by filter)
  base64UrlSafeEncode

charset-decode-filter(used by filter)
  charsetDecode sp quoted-string

count-filter(used by filter)
  count

days-after-now-filter(used by filter)
  daysAfterNow

days-before-now-filter(used by filter)
  daysBeforeNow

first-filter(used by filter)
  first

date-format-filter(used by filter)
  dateFormat sp quoted-string

html-escape-filter(used by filter)
  htmlEscape

html-unescape-filter(used by filter)
  htmlUnescape

jsonpath-filter(used by filter)
  jsonpath sp quoted-string

last-filter(used by filter)
  last

location-filter(used by filter)
  location

nth-filter(used by filter)
  nth sp (integer|placeholder)

regex-filter(used by filter)
  regex sp (quoted-string|regex)

replace-filter(used by filter)
  replace sp quoted-string sp quoted-string

replace-regex-filter(used by filter)
  replaceRegex sp (quoted-string|regex) sp quoted-string

split-filter(used by filter)
  split sp quoted-string

to-date-filter(used by filter)
  toDate sp quoted-string

to-float-filter(used by filter)
  toFloat

to-hex-filter(used by filter)
  toHex

to-int-filter(used by filter)
  toInt

to-string-filter(used by filter)
  toString

url-decode-filter(used by filter)
  urlDecode

url-encode-filter(used by filter)
  urlEncode

url-query-param-filter(used by filter)
  urlQueryParam sp quoted-string

utf8-decode-filter(used by filter)
  utf8Decode

utf8-encode-filter(used by filter)
  utf8Encode

xpath-filter(used by filter)
  xpath sp quoted-string

Lexical Grammar

boolean(used by boolean-option, variable-value, predicate-value, json-value)
  true|false

null(used by variable-value, predicate-value, json-value)
  null

alphanum(used by key-string-text)
  [A-Za-z0-9]

integer(used by integer-option, duration-option, variable-value, nth-filter, float, number)
  digit+

float(used by variable-value, number)
  integer fraction

number(used by greater-or-equal-predicate, greater-predicate, less-or-equal-predicate, less-predicate, predicate-value)
  integer|float

digit(used by json-integer, integer, fraction, exponent)
  [0-9]

hexdigit(used by oneline-hex, unicode-char, json-string-escaped-char)
  [0-9A-Fa-f]

fraction(used by json-number, float)
  . digit+

exponent(used by json-number)
  (e|E) (+|-)? digit+

sp(used by request, response, capture, assert, header-query, certificate-query, cookie-query, xpath-query, jsonpath-query, regex-query, variable-query, predicate, equal-predicate, contain-predicate, end-with-predicate, greater-or-equal-predicate, greater-predicate, include-predicate, less-or-equal-predicate, less-predicate, match-predicate, not-equal-predicate, start-with-predicate, expr, charset-decode-filter, date-format-filter, jsonpath-filter, nth-filter, regex-filter, replace-filter, replace-regex-filter, split-filter, to-date-filter, url-query-param-filter, xpath-filter, lt)
  [\t]

lt(used by hurl-file, request, response, header, body, query-string-params-section, form-params-section, multipart-form-data-section, cookies-section, captures-section, asserts-section, basic-auth-section, options-section, filename-param, capture, assert, option, aws-sigv4-option, ca-certificate-option, client-certificate-option, client-key-option, compressed-option, connect-to-option, connect-timeout-option, delay-option, digest-option, fail-with-body-option, follow-redirect-option, follow-redirect-trusted-option, header-option, http10-option, http11-option, http2-option, http3-option, insecure-option, ipv4-option, ipv6-option, limit-rate-option, max-redirs-option, max-time-option, negotiate-option, netrc-option, netrc-file-option, netrc-optional-option, output-option, path-as-is-option, pinned-public-key-option, proxy-option, resolve-option, repeat-option, retry-option, retry-interval-option, skip-option, unix-socket-option, user-option, variable-option, variables-file-option, verbose-option, very-verbose-option, multiline-string)
  sp* comment? [\n]?

comment(used by lt)
  # ~[\n]*

regex(used by regex-query, match-predicate, regex-filter, replace-regex-filter)
  / regex-content /

regex-content(used by regex)

```
  (regex-text|regex-escaped-char)*

regex-text(used by regex-content)
  ~[\n\/]+

regex-escaped-char(used by regex-content)
  \ ~[\n]
```

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